HL7 VERSION 2.5.1 IMPLEMENTATION GUIDE: ORDERS AND OBSERVATIONS; INTEROPERABLE LABORATORY RESULT REPORTING TO EHR (US REALM), RELEASE 1

ORU^R01

HL7 Version 2.5.1

November, 2007

Chapter Chair:	Hans Buitendijk Siemens Medical Solutions Health Services Corporation
Chapter Chair:	Patrick Loyd Gordon Point Informatics Ltd.
Chapter Chair:	Gunther Schadow, MD Regenstrief Institute for Health Care
Project Chair and Principal Author:	Austin Kreisler SAIC - Science Applications International Corp
Project Chair and Contributing Author:	Steve Steindel Centers for Disease Control and Prevention
Contributing Author	Joann Larson Kaiser Permanente
Technical Writer	J. Paul Harrison SAIC - Science Applications International Corp

Copyright © 2007-2008 Health Level Seven International ® ALL RIGHTS RESERVED. The reproduction of this material in any form is strictly forbidden without the written permission of the publisher. HL7 International and Health Level Seven are registered trademarks of Health Level Seven International. Reg. U.S. Pat & TM Off.

IMPORTANT NOTES:

HL7 licenses its standards and select IP free of charge. **If you did not acquire a free license from HL7 for this document,** you are not authorized to access or make any use of it. To obtain a free license, please visit http://www.HL7.org/implement/standards/index.cfm.

If you are the individual that obtained the license for this HL7 Standard, specification or other freely licensed work (in each and every instance "Specified Material"), the following describes the permitted uses of the Material.

A. HL7 INDIVIDUAL, STUDENT AND HEALTH PROFESSIONAL MEMBERS, who register and agree to the terms of HL7's license, are authorized, without additional charge, to read, and to use Specified Material to develop and sell products and services that implement, but do not directly incorporate, the Specified Material in whole or in part without paying license fees to HL7.

INDIVIDUAL, STUDENT AND HEALTH PROFESSIONAL MEMBERS wishing to incorporate additional items of Special Material in whole or part, into products and services, or to enjoy additional authorizations granted to HL7 ORGANIZATIONAL MEMBERS as noted below, must become ORGANIZATIONAL MEMBERS of HL7.

B. HL7 ORGANIZATION MEMBERS, who register and agree to the terms of HL7's License, are authorized, without additional charge, on a perpetual (except as provided for in the full license terms governing the Material), non-exclusive and worldwide basis, the right to (a) download, copy (for internal purposes only) and share this Material with your employees and consultants for study purposes, and (b) utilize the Material for the purpose of developing, making, having made, using, marketing, importing, offering to sell or license, and selling or licensing, and to otherwise distribute, Compliant Products, in all cases subject to the conditions set forth in this Agreement and any relevant patent and other intellectual property rights of third parties (which may include members of HL7). No other license, sublicense, or other rights of any kind are granted under this Agreement.

C. NON-MEMBERS, who register and agree to the terms of HL7's IP policy for Specified Material, are authorized, without additional charge, to read and use the Specified Material for evaluating whether to implement, or in implementing, the Specified Material, and to use Specified Material to develop and sell products and services that implement, but do not directly incorporate, the Specified Material in whole or in part.

NON-MEMBERS wishing to incorporate additional items of Specified Material in whole or part, into products and services, or to enjoy the additional authorizations granted to HL7 ORGANIZATIONAL MEMBERS, as noted above, must become ORGANIZATIONAL MEMBERS of HL7.

Please see http://www.HL7.org/legal/ippolicy.cfm for the full license terms governing the Material.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

This page intentionally left blank.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE OF CONTENTS

1. INTRODUCTION 1
1.1 Purpose 1
1.2 Audience 1
1.3 Scope 1
1.4 Assumptions 2
1.5 Conventions 2
2. MESSAGING INFRASTRUCTURE 3
2.1 Messaging Framework 3
2.1.1 Delimiters 3
2.1.2 Null Values 4
2.1.3 Lengths 4
2.1.4 Snapshot processing 5
2.2 Use Of Escape Sequences In Text Fields 5
2.3 Data Types 6
2.3.1 Data Type Table Abbreviations 7
2.3.2 CE – Coded Element 8
2.3.3 CQ – Composite Quantity with Units 9
2.3.4 CWE – Coded with Exceptions 9
2.3.5 CX – Extended Composite ID with Check Digit 10
2.3.6 DR – Date/Time Range 11
2.3.7 DT – Date 11
2.3.8 DTM – Date/Time 11
2.3.9 ED – Encapsulated Data 12
2.3.10 EI – Entity Identifier 12
2.3.11 EIP – Entity Identifier Pair 13
2.3.12 FN – Family Name 13
2.3.13 FT – Formatted Text Data 14
2.3.14 HD – Hierarchic Designator 14
2.3.15 ID – Coded Value for HL7-Defined Tables 14
2.3.16 IS – Coded Value for User-Defined Tables 15
2.3.17 MSG – Message Type 15
2.3.18 NM – Numeric 15
2.3.19 PL – Person Location 15
2.3.20 PRL – Parent Result Link 16

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.21 PT – Processing Type *17*
2.3.22 RP – Reference Pointer *17*
2.3.23 SAD – Street Address *19*
2.3.24 SI – Sequence ID *19*
2.3.25 SN – Structured Numeric *20*
2.3.26 ST – String Data *20*
2.3.27 TM – Time *21*
2.3.28 TS – Time Stamp *21*
2.3.29 TX – Text Data *21*
2.3.30 VID – Version Identifier *22*
2.3.31 XAD – Extended Address *22*
2.3.32 XCN – Extended Composite ID Number and Name for Persons *23*
2.3.33 XON – Extended Composite Name and Identification Number for Organizations *24*
2.3.34 XPN – Extended Person Name *25*

3. MESSAGE PROFILE - LABORATORY TO EHR 27
3.1 Use Case Model 27
3.2 Dynamic Interaction Model 28
3.3 Dynamic Definition 29
3.4 Interactions 29

4. MESSAGES 32
4.1 ORU^R01^ORU_R01 33
4.2 ACK^R01^ACK 36

5. SEGMENT AND FIELD DESCRIPTIONS 38
5.1 Segment Attribute Table Abbreviations 38
5.2 MSH – Message Header Segment 40
5.2.1 HL7 Table 0155 – Accept/Application Acknowledgment Conditions (Constrained from the Full HL7 Table) 42
5.3 MSA – Acknowledgement Segment 43
5.3.1 HL7 Table 0008 – Acknowledgement Code (Constrained from the Full HL7 Table) 43
5.4 ERR – Error Segment 44
5.5 PID – Patient Identification Segment 45
5.6 PV1 – Patient Visit Information 48
5.7 PV2 – Patient Visit – Additional Information Segment 51
5.8 ORC – Common Order Segment 55
5.9 OBR – Observation Request Segment 57
5.10 TQ1 – Timing/Quantity Segment 63

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.11 OBX – Observation/Result Segment 64
5.11.1 HL7 Table 0125 – Value Type (Constrained from the Full HL7 Table) 68
5.12 SPM – Specimen Segment 71

6. CODE SYSTEMS AND VALUE SETS 74
6.1 Vocabulary Constraints 74
6.1.1 LOINC 74
6.1.2 SNOMED-CT 75
6.1.3 Unified Code for Units of Measure (UCUM) 75
6.1.4 UB-04 Data Specification Manual 75

7. EXAMPLE LABORATORY RESULT MESSAGES 76
7.1 Minimal Message with Acknowledgement 76
7.1.1 Example: Minimal Message 76
7.1.2 Example: Successful Receipt Message 77
7.1.3 Example: Error on Receipt Message 77
7.1.4 Example: Error on Receipt - Warning 78
7.1.5 Example: Reject Receipt Message 78
7.2 Electrolytes Laboratory Result Message 79
7.2.1 Example: Electrolytes 79
7.3 Creatinine Clearance Laboratory Result Message 80
7.3.1 Example: Creatinine Clearance 80

APPENDIX A. HL7 REPORTING OF CULTURE AND SUSCEPTIBILITIES 82
A.1 Introduction 82
A.2 Template for Culture Results 82
A.2.1 Examples of Culture Results 83
A.3 Template for Culture and Susceptibility Results 84
A.3.1 Examples of Culture and Susceptibility Results 86
A.4 Linking Parent and Child Results 89

APPENDIX B. CLINICAL LABORATORY IMPROVEMENTS AMENDMENT CONSIDERATIONS, US REALM ONLY 90
B.1 Mandatory Reporting Requirements 90
B.2 Report Retention 92
B.3 Authorized Parties 92

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

INDEX OF TABLES

Table 2-1. Delimiters 3
Table 2-2. Data Types 6
Table 2-3. Data Type Attributes 7
Table 2-4. HL7 Component Table – CE – Coded Element 8
Table 2-5. HL7 Component Table – CQ –Composite Quantity with Units 9
Table 2-6. HL7 Component Table – CWE – Coded with Exceptions 9
Table 2-7. HL7 Component Table – CX – Extended Composite ID with Check Digit 10
Table 2-8. HL7 Component Table – DR – Date/Time Range 11
Table 2-9. HL7 Component Table – DT – Date 11
Table 2-10. HL7 Component Table – DTM – Date/Time 11
Table 2-11. HL7 Component Table – ED – Encapsulated Data 12
Table 2-12. HL7 Component Table – EI – Entity Identifier 12
Table 2-13. HL7 Component Table – EIP – Entity Identifier Pair 13
Table 2-14. HL7 Component Table – FN – Family Name 13
Table 2-15. HL7 Component Table – FT – Formatted Text Data 14
Table 2-16. HL7 Component Table – HD – Hierarchic Designator 14
Table 2-17. HL7 Component Table – ID – String Data 14
Table 2-18. HL7 Component Table – IS – String Data 15
Table 2-19. HL7 Component Table – MSG – Message Type 15
Table 2-20. HL7 Component Table – NM – Numeric 15
Table 2-21. HL7 Component Table – PL – Person Location 15
Table 2-22. HL7 Component Table – PRL –Parent Result Link 16
Table 2-23. HL7 Component Table – PT –Processing Type 17
Table 2-24. HL7 Component Table – RP – Reference Pointer 17
Table 2-25. HL7 Table 0834 – MIME Type 19
Table 2-26. HL7 Component Table – SAD – Street Address 19
Table 2-27. HL7 Component Table – SI –Sequence ID 19
Table 2-28. HL7 Component Table –SN – Structured Numeric 20
Table 2-29. HL7 Component Table – ST – String Data 20
Table 2-30. HL7 Component Table – TM – Time 21
Table 2-31. HL7 Component Table – TS – Time Stamp 21
Table 2-32. HL7 Component Table – TX – Text Data 21
Table 2-33. HL7 Component Table – VID – Version Identifier 22
Table 2-34. HL7 Component Table – XAD – Extended Address 22

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Index of Tables

Table 2-35. HL7 Component Table – XCN – Extended Composite ID Number and Name for Persons 23
Table 2-36. HL7 Component Table – XON – Extended Composite Name and Identification Number for Organizations 24
Table 2-37. HL7 Component Table – XPN – Extended Person Name 25
Table 3-1. Use Case: Laboratory to EHR 27
Table 3-2. Dynamic Definition 29
Table 3-3. Interactions 29
Table 4-1. Message Attributes 32
Table 4-2. ORU^R01^ORU_R01 Abstract Message Syntax 33
Table 4-3. ACK^R01^ACK Abstract Message Syntax 36
Table 5-1. Segment Attributes 38
Table 5-2. Message Header Segment (MSH) 40
Table 5-3. HL7 Table 0155 – Accept/Application Acknowledgment Conditions 42
Table 5-4. Message Response Segment (MSA) 43
Table 5-5. HL7 Table 0008 – Acknowledgement Code 43
Table 5-6. Error Segment (ERR) 44
Table 5-7. Patient Identification Segment (PID) 45
Table 5-8. PV1 - Patient Visit Information 48
Table 5-9. PV2 - Patient Visit – Additional Information 51
Table 5-10. Common Order Segment (ORC) 55
Table 5-11. Observation Request Segment (OBR) 57
Table 5-12. Time/Quantity Segment for Order Group 63
Table 5-13. Observation/Result Segment (OBX) 64
Table 5-14. HL7 Table 0125 – Value Type 68
Table 5-15. Specimen Segment (SPM) 71
Table 6–1. Lab LOINC 74
Table 6-2. SNOMED-CT 75
Table B-1. Mandatory Reporting Requirements 90

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

This page intentionally left blank.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

1. Introduction

The *U.S. Realm – Interoperability Specification: Lab Result Message to EHR* is the Health Level Seven (HL7) response to a request from the U.S. Health Information Technology Standards Panel (HITSP) for a standard laboratory message to meet the requirements of its use case. In March, 2006, the HHS Office of the National Coordinator For Health IT published the *Harmonized Use Case for Electronic Health Records (Laboratory Result Reporting)* (Click here to see the use case) in response to a request and specifications from the American Health Information Community. The use case focuses on widely-available, well-standardized methods that will support the secure access to electronic laboratory results and interpretations for clinical care by authorized parties and is driven by the need for timely electronic access to ordered, referred and historical lab results. Ordering clinicians receive lab test results as a response to an order by having the test results sent either directly to the clinician's EHR system (local or remote) or to another clinical data system in support of the provisioning of historical results.

HITSP received the harmonized use case and developed *Interoperability Specification 1 (IS01) for Electronic Health Records Laboratory Reporting* (Click here to see the specification). HITSP has requested this implementation guide to serve as guidance for implementers and those certifying electronic health record systems. This document is intended to meet the needs and requirements of implementation guidance in relevant HITSP Interoperability Specifications.

1.1 PURPOSE

This guide contains the necessary specifications for clinical laboratory results reporting to EHRs for use in the U.S. Realm. It is one component of an overall project for HITSP to establish implementation guides for aspects of the American Health Information Community (AHIC) use cases that can be recognized by the Secretary of the Department of Health and Human Services (HHS). In particular, this guide addresses messaging content and dynamics related to the *AHIC Harmonized Use Case for Electronic Health Records (Laboratory Result Reporting)*, an AHIC EHR-Lab Result use case. While not addressing specific requirements of other HHS use cases, this guide is intended to enable other HHS use cases, such as Biosurveillance, to be met through additional constraints on this guide and/or other profiles.

1.2 AUDIENCE

This guide is designed to be used by analysts and developers who require guidance on optional and ambiguous elements of the HL7 Version 2.5.1 ORU Unsolicited Observation Message relative to the AHIC EHR-Lab Result use case. Users of this guide must be familiar with the details of HL7 message construction and processing. This guide is not intended to be a tutorial on that subject.

1.3 SCOPE

This specification covers the exchange of laboratory results from the testing source between providers and other authorized parties, such as Public Health. Be aware, this guide does not address laboratory orders.

One of the primary features of this implementation guide is its focus on key points of broad interoperability. These key points include the following:

- **Use of strong identifiers for key information objects** – These information objects include patients, orders, providers and organizations. A strong identifier is one that uniquely identifies the object in question in a global fashion. This means the identifier includes enough information to remain unique when taken out of the context within which the identifier was created. This is accomplished through the use of assigning authorities for the identifier. In this guide, an assigning authority is normally handled as an ISO Object Identifier (OID). The combination of the identifier and the OID

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

should always be unique. Assigners of identifiers need to create and register OIDs such that each identifier associated with an assigning authority is globally unique (i.e., it does not collide with another identifier assigned by the same or another assigning authority). This uniqueness ensures that each identifier can be broadly shared among independent healthcare organizations and still point to its originally associated object.

- **Use of Vocabulary Standards** This guide calls for specific vocabulary standards for the exchange of laboratory information. Use of standard vocabularies is important for a number of reasons. Use of standard vocabularies allows broad distribution of healthcare information without the need for individual institutions to exchange master files for data such as test codes, result codes, etc. Each institution maps its own local vocabularies to the standard code, allowing information to be shared broadly, rather than remaining isolated as a single island of information. Standard vocabularies, particularly coded laboratory results, enable more automated decision support for patient healthcare, as well as more automated public health surveillance of populations.

1.4 ASSUMPTIONS

This use case inherits the same assumptions as those identified in the *Harmonized Use Case for Electronic Health Records (Laboratory Results Reporting)* from the Office of the National Coordinator for Health Information Technology (ONC). These assumptions are summarized as follows:

- Infrastructure is in place to allow accurate information exchange between information systems.
- Providers access lab test results through either an EHR or a clinical data system.
- Privacy and security has been implemented at an acceptable level.
- All participants agree to all standards, methodologies, consent, privacy and security.
- Legal and governance issues regarding data access authorizations, data ownership and data use are outside the scope of this document.
- The order, paper or electronic, associated with the lab result contains sufficient information for the laboratory to construct the lab result message properly.

1.5 CONVENTIONS

The following conventions have been used in establishing this guide:

- The rules outlined in *HL7 2.5.1, Chapter 2, Section 2.12, Conformance Using Message Profiles,* were used to document the use case for, and constraints applied to, the messages described in this guide.
- Data types have been described separately from the fields that use the data types. For details regarding data type field lengths, please refer to *Section 2.1.3, Lengths,* in this document.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2. Messaging Infrastructure

2.1 MESSAGING FRAMEWORK

2.1.1 Delimiters

This profile supports the use of the normal HL7 delimiters. It is recommended, but not required, that implementers be able to send messages using the standard HL7 delimiters. Receivers must, however, be capable of receiving any legal delimiters that are sent in a particular message instance.

This table is pre-adopted from the *HL7 Version 2.6*, which offers information regarding Best Practice. The intent has not changed from *Version 2.5.1*. Note that this implementation guide includes additional constraints and explanations for some of the entries.

TABLE 2-1 – DELIMITERS

Delimiter	Suggested Value	Encoding Character Position	Description
Segment Terminator	<cr>	-	Terminates a segment record. This value cannot be changed by implementers. Additional constraints and Explanation: The <cr> denotes the ASCII-013 carriage return character. There is a common misunderstanding that a line feed character, or carriage return followed by a line feed character, is allowed also. Neither HL7 nor this profile allows either of these 2 as part of the segment terminator. Only the ASCII-013 carriage return is allowed.
Field Separator	|	-	Separates two adjacent data fields within a segment. It also separates the segment ID from the first data field in each segment. Additional constraints and Explanation: It is strongly recommended that senders use ASCII-124, the vertical bar (|) character, as the field separator.
Component Separator	^	1	Separates adjacent components of data fields where allowed. Additional constraints and Explanation: It is strongly recommended that senders use ASCII-094, the caret (^) character, as the component separator.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-1 – DELIMITERS			
Delimiter	**Suggested Value**	**Encoding Character Position**	**Description**
Repetition Separator	~	2	Separates multiple occurrences of a field where allowed. Additional constraints and Explanation: It is strongly recommended that senders use ASCII-126, the tilde character (~), as the repetition separator.
Escape Character	\	3	Use the escape character with any field represented by an ST, TX or FT data type, or fwith the data (fourth) component of the ED data type. If no escape characters are used in a message, this character may be omitted. However, it must be present if subcomponents are used in the message. Best practice is always to include this character. Additional constraints and Explanation: It is strongly recommended that senders use ASCII-091, the backslash (\) character, as the escape character.
Subcomponent Separator	&	4	Separates adjacent subcomponents of data fields where allowed. If there are no subcomponents, this character may be omitted. Best practice is always to include this character. Additional constraints and Explanation: It is strongly recommended that senders use ASCII-038, the ampersand (&) character, as the subcomponent separator.

2.1.2 Null Values

In HL7, a null value is indicated by paired double quotes (|""|) for a field. A null value indicates that the receiver of the message should delete the corresponding piece of information from the data store. For this implementation guide, null values within components and subcomponents are meaningless. For example, |lastname^firstname^""^^^^L| would be interpreted exactly as |lastname^firstname^^^^^L|. The components and subcomponents of a data type constitute a snapshot of the data. The set of data represented by the data type is handled as a complete set; therefore, using the null value to indicate a missing component or subcomponent is unnecessary.

2.1.3 Lengths

In *HL7 Version 2.5*, HL7 assigned lengths to the components of data types, but did not standardize the lengths of the fields that use those data types. This guide employs the following rules for lengths:

- Only lengths of atomic data types are documented. This rule applies to both complex data types and fields in segments.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

- The lengths of atomic data types are based upon either the 2.5.1 length, a length pre-adopted from 2.6, or a length the project team felt was needed to meet the use case.

- Lengths of optional and unsupported segment fields and data type components are outside the scope of this document and, therefore, not specified in this guide.

> *Note: In* HL7 Version 2.5.1*, the length of 65536 has a special meaning: For HL7, "If the maximum length needs to convey the notion of a Very Large Number, the number 65536 should be displayed to alert the user."*
> *In this implementation guide, fields or components with length 65536 should be understood as having no prescribed length. Receivers should be prepared to accept any size chunk of data carried in the field or component.*

2.1.4 Snapshot processing

HL7 distinguishes between two methods of update: the "snapshot" and the "action code/unique identifier" modes. Both modes apply to repeating segments and repeating segment groups. For repeating fields, only snapshot processing applies. For the purpose of this guide, only snapshot processing is supported for segments, segment groups and fields.

2.1.4.0 Repeating Segments

HL7 defines snapshot processing for segments as follows:

> In the "snapshot" mode, the information contained in the set of repeating segments or segment groups from the incoming message replaces the corresponding information in the receiving application. This is equivalent to a deletion of the prior information followed by the addition of the newly supplied information. In this mode, everything (all repeating segments and segment groups) must be sent with every subsequent message in the series of messages.-There is no other way to indicate which ones changed and which ones did not.
> To specify "delete all of the segments in this repeating group" in the snapshot mode, send a single segment with "delete data" (indicated by a value of "") in all fields. This actively signals the receiver that there is information that needs to be deleted. If no segment were sent, this would equate to "no information." No information should not signal the receiver to take an action. There would be risk that the receiver might misinterpret the sender's intent.[1]

2.1.4.1 Repeating Fields

Snapshot processing for repeating fields requires sending a full list of repetitions for each transaction. If the intent is to delete an element, the element is left off the list. This is analogous to the snapshot mode for repeating segments and segment groups. To delete the whole list, transmit the field once with a "Null" in the first component.

Repetitions of fields shall not have empty repetitions followed by repetitions containing data, except where the HL7 standard clearly reserves certain repetitions for specific purposes. For instance, PID-5 Patient Name is a repeating field, the first repetition of which is reserved by HL7 for the legal name. In the case where a name is known for the patient, but is not the legal name, format the name field as follows: |~lastname^firstname^mi^^^^A|.

2.2 USE OF ESCAPE SEQUENCES IN TEXT FIELDS

Senders and receivers using this profile shall handle escape sequence processing as described in *HL7 Version 2.5.1, Chapter 2,Section 2.7.4 (Special Characters*) and *Section 2.7.5 (Hexadecimal)*. Implementers may support escape sequences described in *Sections 2.7.2 (Escape sequences supporting multiple character sets), 2.7.3 (Highlighting), 2.7.6 (Formatted Text)* and *2.7.7 (Local)*.

[1] Taken from HL:7 2.6, Chapter 2, section 2.10.4.1.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3 DATA TYPES

TABLE 2-2 – DATA TYPES		
Data type	Data Type Name	Length
CE	Coded element	
CQ	Composite Quantity with Units	
CWE	Coded with Exceptions	
CX	Extended Composite ID with Check Digit	
DR	Date/Time Range	
DT	Date	8
DTM	Date/Time	24
ED	Encapsulated Data	
EI	Entity Identifier	
EIP	Entity Identifier Pair	
FN	Family Name	
FT	Formatted Text Data	65536
HD	Hierarchic Designator	
ID	Coded Values for HL7 Tables	Variable
IS	Coded value for User-Defined Tables	20
MSG	Message Type	
NM	Numeric	16
PL	Person Location	
PRL	Parent Result Link	
PT	Processing Type	
RP	Reference Pointer	
SAD	Street Address	
SI	Sequence ID	4
SN	Structured Numeric	
ST*	String	999
TM	Time	16
TS	Time Stamp	
TX	Text Data	65536
VID	Version Identifier	

* Indicates that the length is based on a pre-adoption of the H7 Version 2.6 standard.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-2 – DATA TYPES		
Data type	**Data Type Name**	**Length**
XAD	Extended Address	
XCN	Extended Composite ID Number and Name	
XON	Extended Composite Name and ID Number for Organizations	
XPN	Extended Person Name	

2.3.1 Data Type Table Abbreviations

The following sections detail the structure of each data type, including segment name, usage, value set and description. The table below details the abbreviated terms used in the data type tables, as well as the definitions associated with those abbreviations.

TABLE 2-3 – DATA TYPE ATTRIBUTES	
Abbreviation	**Definition**
Seq	Sequence of the elements as numbered in the HL7 segment.
Len	Maximum length of the element. Lengths are only provided for atomic data types. Lengths should be considered recommendations, not absolutes. The receiver can truncate fields, components and sub-components that are longer than the recommended length. The receiver should continue to process a message even when a field, component, or sub-component length exceeds the maximum recommended length identified in this specification.
DT	Data type used by this profile for HL7 element.
Usage	Usage of the field for this profile. Indicates whether the field, component, or subcomponent is required, optional, or conditional in the corresponding segment, field, or component. Legal values are: • R – Required. Must always be populated. • RE – Required, but can be empty. • O – Optional. May optionally be populated. • CE – Conditional. Populated under specified conditions. • C – Conditional, but may be empty. • X – Not used for this profile.
Value Set	The set of coded values to be used with the field. Note: Where a table constraint is indicated, or where HL7 Version 2.6 standards are pre-adopted, the constrained or specified HL7 table is included below the data type table.
Component Name	HL7 descriptor of the element in the data type.
Description/Comments	Context and usage for the element.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Note: In the tables throughout this document, Yellow = This Interoperability Specification does not support the use of this item.

2.3.2 CE – Coded Element

TABLE 2-4 – CODED ELEMENT (CE)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	RE		Identifier	
2	999	ST	RE		Text	It is strongly recommended that text be sent to accompany any identifier. When a coded value is not known, text can still be sent, in which case no coding system should be identified.
3	20	ID	CE	HL70396	Name of Coding System	Condition rule: Required if an identifier is provided in component 1.
4	999	ST	RE		Alternate Identifier	The alternate identifier (from the alternate coding system) should be the closest match for the identifier found in component 1.
5	999	ST	RE		Alternate Text	It is strongly recommended that alternate text be sent to accompany any alternate identifier.
6	20	ID	CE	HL70396	Name of Alternate Coding System	Condition rule: Required if an alternate identifier is provided in component 4.

Usage: The CE data type is used where it is necessary to communicate a code, text and the coding system the code was drawn from. It also allows the communication of an alternate code drawn from another coding system. Many coded fields in this specification identify coding systems or value sets which must be used for the field. When populating the CE data types with these values, this guide does not give preference to the triplet in which the standard code should appear. The receiver is expected to examine the coding system names in components 3 and 6 to determine if it recognizes the coding system.

This guide conforms to the HL7 Version 2.6 convention of using the CWE data type in place of the CE to allow transmission of version information.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.3 CQ – Composite Quantity with Units

TABLE 2-5 - COMPOSITE QUANTITY WITH UNITS (CQ)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	16	NM	R		Quantity	
2		CWE	RE	UCUM	Units	Units of measure must be drawn from the UCUM coding system.

2.3.4 CWE – Coded with Exceptions

TABLE 2-6 - CODED WITH EXCEPTIONS (CWE)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	RE		Identifier	
2	999	ST	RE		Text	It is strongly recommended that text be sent to accompany any identifier. When a coded value is not known, the original text attribute is used to carry the text, not the text component.
3	20	ID	CE	HL70396	Name of Coding System	Condition rule: Required if an identifier is provided in component 1.
4	999	ST	RE		Alternate Identifier	The alternate identifier (from the alternate coding system) should be the closest match for the identifier found in component 1.
5	999	ST	RE		Alternate Text	It is strongly recommended that alternate text be sent to accompany any alternate identifier.
6	20	ID	CE	HL70396	Name of Alternate Coding System	Condition rule: Required if an alternate identifier is provided in component 4.
7	999	ST	RE		Coding System Version ID	
8	999	ST	RE		Alternate Coding System Version ID	
9	999	ST	RE		Original Text	Original Text is used to convey either the text which was the basis for coding, or when there is no code to be sent, only free text.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Usage: The CWE data type is used where it is necessary to communicate a code, text, coding system and the version of coding system the code was drawn from. It also allows the communication of an alternate code drawn from another coding system. Many coded fields in this specification identify coding systems or value sets that must be used for the field. When populating the CWE data types with these values, this guide does not give preference to the triplet in which the standard code should appear. The receiver is expected to examine the coding system names in components 3 and 6 to determine if it recognizes the coding system.

The CWE data type allows communication of an early form of what has come to be called "null flavors." HL7 2.5.1 refers to these as CWE Statuses, where the values are drawn from HL7 table 0353. Support for the CWE Statuses are optional.

This guide conforms to the HL7 Version 2.6 convention of using the CWE data type in place of the CE to allow transmission of version information.

2.3.5 CX – Extended Composite ID with Check Digit

TABLE 2-7 – EXTENDED COMPOSITE ID WITH CHECK DIGIT (CX)

SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	R		ID Number	The ID Number component combined with the Assigning Authority component must uniquely identify the associated object, i.e., whatever object with which the field is associated.
2		ST	O		Check Digit	
3		ID	O		Check Digit Scheme	
4		HD	R		Assigning Authority	
5	5	ID	R	HL70203	Identifier Type Code	
6		HD	O		Assigning Facility	
7		DT	O		Effective Date	
8		DT	O		Expiration Date	
9		CWE	O		Assigning Jurisdiction	
10		CWE	O		Assigning Agency or Department	

Usage: The CX data type is used to carry identifiers. This guide requires that all identifiers be accompanied by assigning authorities, and that all identifiers carry an identifier type. This method allows the exchange of unique identifiers for the associated object across organizational and enterprise boundaries, enabling broad interoperability.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Although the Identifier Type Code component is required, it is not a part of the actual identifier. Rather, it is metadata about the identifier. The ID Number and Assigning Authority component, together, constitute the actual identifier. The reason for this requirement is to promote forward compatibility with HL7 Version 3 identifiers, where there is no concept of identifier type codes. Although this guide does not deal directly with Version 3 constructs, it is intended to work within the context of the HITSP Interoperability constructs, which work with both Version 2.x messaging and Version 3 constructs.

2.3.6 DR – Date/Time Range

TABLE 2-8 – DATE/TIME RANGE (DR)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		TS	RE		Range Start Date/Time	
2		TS	RE		Range End Date/Time	

2.3.7 DT – Date

TABLE 2-9 – DATE (DT)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	8		R		Date	Format: YYYY[MM[DD]]

2.3.8 DTM – Date/Time

TABLE 2-10 – DATE/TIME (DTM)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	24		R		Date/Time	Format: YYYY[MM[DD[HH[MM[SS[.S[S[S[S]]]]]]]]][+/-ZZZZ]

Usage: It is strongly recommended that the time zone offset always be included in the DTM. Specific fields in this implementation guide may require Date/Time to a specific level of granularity, which may require the time zone offset.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.9 ED – Encapsulated Data

TABLE 2-11 - ENCAPSULATED DATA (ED)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		HD	RE		Source Application	Identifier of the application that is the source of the encapsulated data.
2	9	ID	R	HL70191	Type of Data	Identifier of the type of data found in component 5.
3	18	ID	RE	HL70291	Data Subtype	Identifier of the subtype of data found in component 5
4	6	ID	R	HL70299	Encoding	Identifier of the type of encoding to be performed in the data component
5	65536	TX	R		Data	The data in this component must be properly escaped after encoding. Receivers will need to de-escape the text prior to de-encoding.

2.3.10 EI – Entity Identifier

TABLE 2-12 - ENTITY IDENTIFIER (EI)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	R		Entity Identifier	
2	20	IS	O		Namespace ID	
3	999	ST	R		Universal ID	Must be an OID
4	6	ID	R	HL70301	Universal ID Type	Constrained to the value "ISO."

Usage: The EI data type is used to carry identifiers. This guide requires that all entity identifiers be accompanied by assigning authorities. This allows the exchange of unique identifiers for the associated object across organizational and enterprise boundaries, enabling broad interoperability.

In the EI data type, the Namespace ID, Universal ID and Universal ID type correspond with the HD data type identified elsewhere. These types together are commonly considered the assigning authority for the identifier. The Entity Identifier and assigning authority components, together, constitute the actual identifier. This method promotes forward compatibility with HL7 Version 3 identifiers, where there is no concept of identifier type codes. Although this guide does not deal directly with Version 3 constructs, it is

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

intended to work within the context of the HITSP Interoperability constructs, which deal with both Version 2.x messaging and Version 3 constructs.

2.3.11 EIP – Entity Identifier Pair

TABLE 2-13 – ENTITY IDENTIFIER PAIR (EIP)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		EI	RE		Placer Assigned Identifier	
2		EI	CE		Filler Assigned Identifier	Condition rule: Component 2 will be required if the field is OBR-29; otherwise, the component is RE. This is necessary to accommodate the use of EIP in SPM-2

2.3.12 FN – Family Name

TABLE 2-14 – FAMILY NAME (FN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	R		Surname	
2		ST	O		Own Surname Prefix	
3		ST	O		Own Surname	
		ST	O		Surname Prefix From Partner/Spouse	
5		ST	O		Surname From Partner/Spouse	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.13 FT – Formatted Text Data

TABLE 2-15 - FORMATTED TEXT DATA (FT)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
	65536		R		Coded Value for HL7-Defined Tables	

Usage: The FT data type allows use of the formatting escape sequences documented in *HL7 Version 2.5.1, Chapter 2, Section 2.7 - Use of Escape Sequences in Test Fields*.

2.3.14 HD – Hierarchic Designator

TABLE 2-16 - HIERARCHIC DESIGNATOR (HD)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		IS	O		Namespace ID	
2	999	ST	R		Universal ID	Must be an OID.
3	6	ID	R	HL70301	Universal ID Type	Constrained to the value 'ISO'.

Usage: The HD data type is used directly to identify objects such as applications or facilities. It is also used as a component of other data types where it is typically an assigning authority for an identifier. It may be used to identify a URI. Where this capability is used in this specification, that usage is described separately. Note that the HD data type has been constrained to carry an OID identifying an application, a facility, or an assigning authority.

2.3.15 ID – Coded Value for HL7-Defined Tables

TABLE 2-17 - CODED VALUE FOR HL7-DEFINED TABLES (ID)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	Variable		R		Coded Value for HL7-Defined Tables	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.16 IS – Coded Value for User-Defined Tables

TABLE 2-18 - CODED VALUE FOR USER-DEFINED TABLES (ID)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	20		R		Coded Value for User-Defined Tables	

2.3.17 MSG – Message Type

TABLE 2-19 - MESSAGE TYPE (MSG)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	3	ID	R	HL70076	Message Code	
2	3	ID	R	HL7003	Trigger Event	
3	7	ID	R	HL70354	Message Structure	

2.3.18 NM – Numeric

TABLE 2-20 - NUMERIC (NM)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	16		R		Numeric	

2.3.19 PL – Person Location

TABLE 2-21 - PERSON LOCATION (PL)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		IS	O	HL70302	Point of Care	
2		IS	O	HL70303	Room	
3		IS	O	HL70304	Bed	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-21 – PERSON LOCATION (PL)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
4		HD	O		Facility	
5		IS	O		Location Status	
6	20	IS	RE	HL70305	Person Location Type	
7		IS	O		Building	
8		IS	O		Floor	
9		ST	O		Location Description	
10		EI	O		Comprehensive Location Identifier	
11		HD	O		Assigning Authority for Location	

2.3.20 PRL – Parent Result Link

TABLE 2-22 – PARENT RESULT LINK (PRL)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		CWE	R		Parent Observation Identifier	Identifier of the OBX-3 Observation ID of the parent result. Typically, this is used in microbiology results where the sensitivities are linked to the specific culture OBX where the organism was identified.
2	999	ST	RE		Parent Observation Sub-Identifier	Identifier of the OBX-4 Observation Sub-ID associated with the OBX-3 Observation ID of the parent result. Typically, this is used in microbiology results where the sensitivities are linked to the specific culture OBX where the organism was identified. The combination of OBX-3 and OBX-4 must be unique within a particular OBR.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-22 – PARENT RESULT LINK (PRL)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
3	65536	TX	RE		Parent Observation Value Descriptor	Taken from the OBX-5 of the parent result. If the OBX-5 is coded data, this would be the value of the text component of the CE or CWE data type. Length of PRL-3 is based on pre-adoption of HL7 Version 2.6 standards.

Usage: See *Appendix A*, of this document for details on how this data type and the EIP data type are used in parent/child result linking. Use of data type CWE for sequence 1 reflects a pre-adoption of HL7 Version 2.6 standards.

2.3.21 PT – Processing Type

TABLE 2-23 – PROCESSING TYPE (PT)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	1	ID	R	HL70103	Processing ID	
2		ID	O		Processing Mode	

2.3.22 RP – Reference Pointer

TABLE 2-24 – REFERENCE POINTER (RP)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	R		Pointer	Pointer to the object. For URIs, it contains the path and query parts. Example: /library/committees/orders/minutes/OO_Minutes_20100101.pdf Length of RP-1 is based on pre-adoption of HL7 Version 2.6 standards.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-24 – REFERENCE POINTER (RP)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
2		HD	R		Application ID	Unique identifier of the application that holds the object being pointed to. For URIs, it contains the scheme and authority parts.
2.1	20	IS	O		Namespace ID	
2.2	999	ST	R		Universal ID	Field that, for URIs, contains the scheme and authority parts. Example: http://www.hl7.org Length of RP-2.2 is based on pre-adoption of HL7 Version 2.6 standards.
2.3	6	ID	R	HL7301	Universal ID Type	Field that, for URIs, contains the literal value 'URI'.
3	11	ID	RE	HL70834	Type of Data	Identifier of the type of data pointed to. For the URI example above, this is "application." Length of RP-3 is based on pre-adoption of HL7 Version 2.6 standards.
4	32	ID	RE	HL70291	Subtype	Identifier of the subtype of data pointed to. For the URI example above, this is ".pdf," indicating portable document format. Length of RP-4 is based on pre-adoption of HL7 Version 2.6 standards.

Usage: The field uses the RP data type to allow communication of pointers to images, sound clips, XML documents, HTML markup, etc. The RP data type is used when the object being pointed to is too large to transmit directly.

This specification defines the mechanism for exchanging pointers to objects, but does not address the details of applications actually accessing and retrieving the objects over a network.

The most common scheme for passing a pointer is to use a Universal Resource Identifier (see `http://ietf.org/rfc/rfc2396.txt` for a detailed definition). The general format of a URI is in the form <scheme>://<authority><path>?<query>. The scheme and authority portions appear in the Application ID component, Universal ID subcomponent. The path and query portion of the URI appear in the Pointer component of the RP data type.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.22.0 HL7 Table 0834 – MIME Type

TABLE 2-25 - HL7 TABLE 0834 - MIME TYPE			
Value	Description	Usage	Comments
application	Application data	O	
audio	Audio data	R	
image	Image data	R	
model	Model data	O	
text	Text data	R	
video	Video data	R	
multipart	MIME multipart package	O	

2.3.23 SAD – Street Address

TABLE 2-26 - STREET ADDRESS (SAD)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	R		Street or Mailing Address	
2		ST	O		Street Name	
3		ST	O		Dwelling Number	

Usage: The SAD is used only as a component of the XAD data type.

2.3.24 SI – Sequence ID

TABLE 2-27 - SEQUENCE ID (SI)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	4		R		Sequence ID	Non-negative integer up to 9999. May be further constrained to limit the number of times a segment may repeat.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.25 SN – Structured Numeric

TABLE 2-28 – STRUCTURED NUMERIC (SN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	2•	ST	RE		Comparator	Component that must be one of ">" or "<" or ">=" or "<=" or "=" or "<>". This component defaults to "=" if empty.
2	16	NM	RE		Num1	
3	1•	ST	RE		Separator/Suffix	Component that must be one of "-" or "+" or "/" or "." or ":".
4	16	NM	RE		Num2	

Usage: The SN data type carries a structured numeric result value. Structured numeric values include intervals (^0^-^1), ratios (^1^/^2 or ^1^:^2), inequalities (<^10), or categorical results (2^+).

2.3.26 ST – String Data

TABLE 2-29 – STRING DATA (ST)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999		R		String Data	

Usage: The ST data type is normally used for short text strings. No leading blanks (space characters) are permitted. Trailing blanks are permitted.

• Length constrained to size of data for field.

• Length constrained to size of data for field.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.27 TM – Time

TABLE 2-30 – TIME (TM)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	16		R		Time	Format: HH[MM[SS[.S[S[S[S]]]]]][+/-ZZZZ]

Usage: It is strongly recommended that the time zone offset always be included in the TM. Specific fields in this implementation guide may require time to a specific level of granularity, which may require the time zone offset.

2.3.28 TS – Time Stamp

TABLE 2-31 – TIME STAMP(TS)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	24	DTM	R		Time	
2		ID	X		Degree of Precision	Deprecated as of *HL7 Version 2.3*. See component 1 (DTM) for the current method of designating degree of precision.

2.3.29 TX – Text Data

TABLE 2-32 – TEXT DATA (TX)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	65536		R		Text Data	

Usage: The TX data type is used to carry string data intended for display purposes. It can contain leading blanks (space characters).

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

2.3.30 VID – Version Identifier

TABLE 2-33 – VERSION IDENTIFIER (VID)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	5	ID	R	HL70104	Version ID	Restricted to *2.5.1* in this guide.
2		CWE	O		Internationalization Code	
3		CWE	O		International Version ID	

2.3.31 XAD – Extended Address

TABLE 2-34 – EXTENDED ADDRESS (XAD)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		SAD	RE		Street Address	
2	999	ST	RE		Other Designation	
3	999	ST	RE		City	
4	999	ST	RE		State or Province	
5	12•	ST	RE		Zip or Postal Code	
6	3	ID	RE	ISO 3166-1	Country	
7	3	ID	RE	HL70190	Address Type	
8		ST	O		Other Geographic Designation	
9	20	IS	RE	289	County/Parish Code	No standard value set is used. FIPS codes may be used.
10		IS	O		Census Tract	
11		ID	O		Address Representation Code	
12		DR	X		Address Validity Range	Deprecated as of *HL7 Version 2.5*. See XAD-13 Effective Date and XAD-14 Expiration Date components.
13		TS	O		Effective Date	

• Length constrained to HL7 length from data type.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-34 – EXTENDED ADDRESS (XAD)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
14		TS	O		Expiration Date	

2.3.32 XCN – Extended Composite ID Number and Name for Persons

TABLE 2-35 – EXTENDED COMPOSITE ID NUMBER AND NAME FOR PERSONS (XCN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	RE		ID Number	If this is a person, the ID must be accompanied by an OID for the assigning authority (component 9). If this is an organization, the OID for the organization should be provided here. Length of XCN-1 is based on pre-adoption of *HL7 Version 2.6* standards.
2		FN	RE		Family Name	
3	999	ST	RE		Given Name	
4	999	ST	RE		Second and Further Given Names or Initials Thereof	
5	999	ST	RE		Suffix (e.g., JR or III)	
6	999	ST	RE		Prefix (e.g., DR)	
7		IS	O		Degree (e.g., MD)	
8		IS	O	HL70297	Source Table	
9		HD	CE		Assigning Authority	Condition rule: Required if component 1 (ID Number) is populated.
10	1	ID	RE	HL70200	Name Type Code	
11		ST	O		Identifier Check Digit	
12	3	ID	CE	HL70061	Check Digit Scheme	Condition rule: Required if component 11 (Identifier Check Digit) is populated.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-35 – EXTENDED COMPOSITE ID NUMBER AND NAME FOR PERSONS (XCN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
13	5	ID	RE	HL70203	Identifier Type Code	
14		HD	O		Assigning Facility	
15		ID	O		Name Representation Code	
16		CWE	O		Name Context	
17		DR	X		Name Validity Range	Deprecated as of *HL7 Version 2.5*. See XCN-19 Effective Date and XCN-20 Expiration Date components.
18		ID	O		Name Assembly Order	
19		TS	O		Effective Date	
20		TS	O		Expiration Date	
21	999	ST	RE		Professional Suffix	
22		CWE	O		Assigning Jurisdiction	
23		CWE	O		Assigning Agency or Department	

2.3.33 XON – Extended Composite Name and Identification Number for Organizations

TABLE 2-36 – EXTENDED COMPOSITE NAME AND IDENTIFICATION NUMBER FOR ORGANIZATIONS (XON)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1	999	ST	RE		Organization Name	
2	20	IS	RE	HL70204	Organization Name Type Code	
3		NM	X		ID Number	(Deprecated as of *HL7 Version 2.5*.) Use XON-10 Organization Identifier.
4		NM	O		Check Digit	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-36 – EXTENDED COMPOSITE NAME AND IDENTIFICATION NUMBER FOR ORGANIZATIONS (XON)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
5		ID	O		Check Digit Scheme	
6		HD	CE		Assigning Authority	Condition rule: Required if component 10 (Organization Identifier) is populated.
7	5	ID	RE	HL70203	Identifier Type Code	
8		HD	O		Assigning Facility	
9		ID	O		Name Representation Code	
10	999	ST	RE		Organization Identifier	Length of XON-10 is based on pre-adoption of HL7 Version 2.6 standards.

2.3.34 XPN – Extended Person Name

TABLE 2-37 – EXTENDED PERSON NAME (XPN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
1		FN	RE		Family Name	
2	999	ST	RE		Given Name	
3	999	ST	RE		Second and Further Given Names or Initials Thereof	
4	999	ST	RE		Suffix (e.g., JR or III)	
5	999	ST	RE		Prefix (e.g., DR)	
6		IS	O		Degree (e.g., MD)	
7	1	ID	RE	HL70200	Name Type Code	
8		ID	O		Name Representation Code	
9		CWE	O		Name Context	
10		DR	X		Name Validity Range	Deprecated as of *HL7 Version 2.5*. See XPN-12 Effective Date and XPN-13 Expiration Date components.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 2-37 – EXTENDED PERSON NAME (XPN)						
SEQ	LEN	DT	Usage	Value Set	Component Name	Comments
11		ID	O		Name Assembly Order	
12		TS	O		Effective Date	
13		TS	O		Expiration Date	
14	999	ST	RE		Professional Suffix	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

3. Message Profile - Laboratory to EHR

3.1 USE CASE MODEL

TABLE 3-1 – USE CASE LABORATORY TO EHR	
Description	The *Laboratory Messaging to EHR Use Case* is a subset of the *Harmonized Use Case for Electronic Health Records (Laboratory Result Reporting)* published by the Office of the National Coordinator for Health Information Technology (ONC). This document focuses on the subset of the use case that applies to the exchange of laboratory results between the laboratory and EHR. It does not cover querying patient demographics or laboratory results. It does include acknowledgments of receipt of transactions.
Actors	• Laboratory Result Sender – The laboratory result sender actor is an application capable of performing laboratory testing on specimens. The laboratory application is capable of transmitting the results of laboratory testing to a receiver. In the use case, the laboratory result sender is identified as a "Laboratory Organization." • Laboratory Result Receiver – The laboratory result receiver is an application capable of receiving results of laboratory testing. Typically this actor represents an EHR application. The laboratory result receiver may be associated with the ordering provider or another provider, commonly referred to as a "copy-to provider," that needs to have access to the results. In the use case, the laboratory result receiver is identified as either the "Clinician" or "Data Repository."
Assumptions	This use case inherits the same assumptions as those identified in the *Harmonized Use Case for Electronic Health Records (Laboratory Results Reporting)* published by ONC. These assumptions are summarized as follows: • Infrastructure is in place to allow correct information exchange between information systems. • Providers access lab test results either through an EHR or a clinical data system. • Privacy and security has been implemented at an acceptable level. • All participants agree to all standards, methodologies, consent, privacy and security. • Legal and governance issues regarding data access authorizations, data ownership and data use are outside the scope of this document. The following are preconditions[2] for the use of this profile: • The order contains the unambiguous names and electronic addresses for the other authorized providers of care. • When needed, the patient is registered in a Patient ID Cross-Referencing system that includes

[2] From *HITSP Interoperability Specification: Send Laboratory Result Message to Ordering Clinician and Providers of Care Transaction Package*, dated

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

	both the laboratory patient ID and the clinician's patient ID. • For the electronic laboratory result, the laboratory has transformed any local codes into HITSP-specified terminologies before transmission. Additional Preconditions: • A valid order for laboratory testing exists.

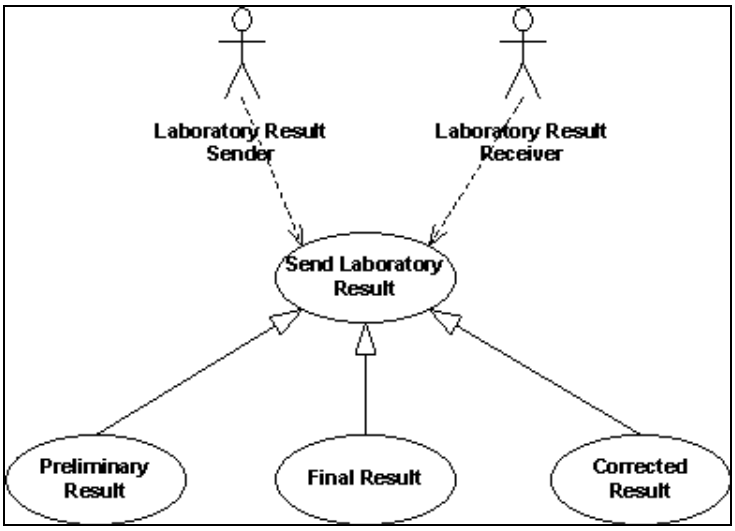

Figure 3-1. Send Laboratory Result Use Case Model

3.2 DYNAMIC INTERACTION MODEL

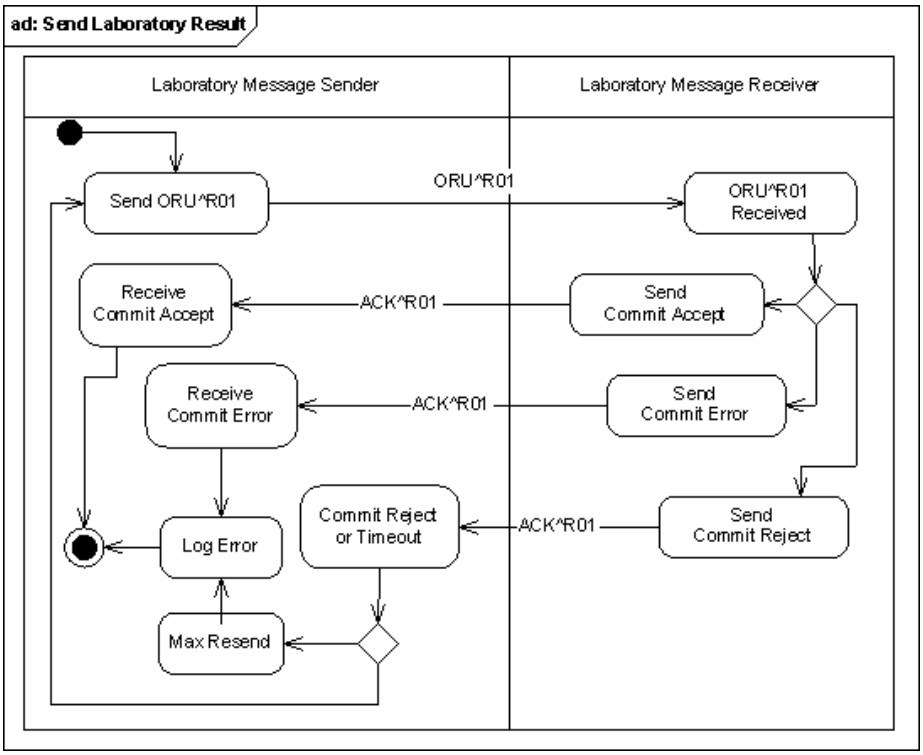

Figure 3-2. Activity Diagram for Send Laboratory Result Use Case

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

3.3 DYNAMIC DEFINITION

TABLE 3-2 – DYNAMIC DEFINITION	
Item	**Value**
Profile ID	USLabReport
HL7 Version	2.5.1
Accept Acknowledgement	AL – Always
Application Acknowledgement	Refer to HL7 Table 0155 – Accept/Application Acknowledgment conditions in section 5.2.1 for valid values.
Acknowledgement Mode	Immediate
Profile Type	Realm Constrainable Profile
Message Types	ORU^R01^ORU_R01, ACK^R01^ACK
Encoding	ER7 (required) 2.5.1 XML (optional)

3.4 INTERACTIONS

TABLE 3-3 – INTERACTIONS							
Event	**Description**	**Usage**	**When Used**	**Message Type**	**Receiver Action**	**Sender**	**Data Values**
Order Received, No specimen	Order received; specimen not yet received	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=O
Specimen Received	No results available; specimen received, procedure incomplete	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=I
Procedure Scheduled	No results available; procedure scheduled, but not done	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=S
Preliminary Result	Preliminary: A verified early result is available, final results not yet	R	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=P

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 3-3 – INTERACTIONS							
Event	Description	Usage	When Used	Message Type	Receiver Action	Sender	Data Values
	obtained						
Partial Result	Some, but not all, results available	O	Some Final Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=A
Unverified Result	Results stored; not yet verified	O	Preliminary Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=R
Final Result	Final results; results stored and verified. Can only be changed with a corrected result.	R	Final Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=F
Correction	Correction to results	R	Corrected Result	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=C
Testing Not Done	No results available; Order canceled.	O	Cancelled Test	ORU^R01^ ORU_R01	Commit Accept, Commit Reject or Commit Error	Laboratory Result Sender	ORC-1=RE OBR-25=X
No Order	No order on record for this test. (Used only on queries)	X	-	varies	NA	Laboratory Result Sender	ORC-1=RE OBR-25=Y
No Patient Record	No record of this patient. (Used only on queries)	X	-	varies	NA	Laboratory Result Sender	ORC-1=RE OBR-25=Z
Commit Accept	Enhanced mode: Accept acknowledgment : Commit Accept	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CA
Commit Error	Enhanced mode: Accept acknowledgment : Commit Error	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CE
Commit Reject	Enhanced mode: Accept acknowledgment : Commit Reject	R	All Cases	ACK^R01^ ACK	None	Laboratory Result Receiver	MSA-1=CR

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

4. Messages

The following sections detail the structure of each message, including segment name, usage, cardinality and description. The table below details the abbreviated terms used in the message tables, as well as the definitions associated with those abbreviations.

TABLE 4 –1 – MESSAGE ATTRIBUTES	
Abbreviation	**Definition**
Segment	Three-character code for the segment and the abstract syntax (*e.g.*, the square and curly braces). • [XXX] Optional • { XXX } Repeating • XXX Required • [{ XXX }] Optional and Repeating Note that for segment groups there is no segment code present, but the square and curly braces will still be present.
Name	Name of the segment or segment group element.
Usage	Use of the segment for this guide. Indicates if the segment is required, optional, or conditional in a message. Legal values are: • R – Required. Must always be populated. • RE – Required, but can be empty. • O – Optional. • C – Conditional. Must be populated based on computable Conditionality Statement. • CE – Conditional, but can be empty. • X – Not used.
Cardinality	Minimum and maximum number of times the element may appear. • [0..0] Element never present. • [0..1] Element may be omitted and can have, at most, one occurrence. • [1..1] Element must have exactly one occurrence. • [0..n] Element may be omitted or may repeat up to *n* times. • [1..n] Element must appear at least once, and may repeat up to *n* times. • [0..*] Element may be omitted or repeat an unlimited number of times. • [1..*] Element must appear at least once, and may repeat unlimited number of times. • [m..n] Element must appear at least *m*, and at most, *n* times.
Section	The part of this guide that describes the segment.
Description	A short description of the use of the segment.

Note: In the tables throughout this document, Yellow = This Interoperability Specification does not support the use of this item.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

4.1 ORU^R01^ORU_R01

The ORU^R01 message is constrained for transmitting laboratory results from the testing source between providers and other authorized parties, such as Public Health.

TABLE 4-2 – ORU^R01^ORU^R01 ABSTRACT MESSAGE SYNTAX					
Segment	**Name**	**Usage**	**Cardinality**	**Section**	**Description**
	Header Begin				
MSH	Message Header	R	[1..1]	5.2	The message header (MSH) segment contains information describing how to parse and process the message. This includes identification of message delimiters, sender, receiver, message type, timestamp, etc.
[{SFT}]	Software Segment	O	[0..*]		
{	*PATIENT_RESULT Begin*	R	[1..*]		The patient result group has been constrained to support at least one patient result.
[	*PATIENT Begin*	RE	[0..1]		The patient group may be empty in this message type. To comply with CLIA regulations for testing of human specimens, as outlined in *Appendix B*, the patient group needs to be sent.
PID	Patient Identification	R	[1..1]	5.2.1	The patient identification (PID) segment is used to provide basic demographics regarding the subject of the testing. The subject may be a person, animal, place, or thing.
[PD1]	Additional Demographics	O	[0..1]		
[{NTE}]	Notes and Comments for PID	RE	[0..1]		
[{NK1}]	Next of Kin/Associated Parties	O	[0..*]		
[	*VISIT Begin*	RE	[0..1]		The visit group may be empty in this message type.
PV1	Patient Visit	R	[0..1]		
[PV2]	Patient Visit – Additional Information	RE	[0..1]		
]	*VISIT End*				
]	*PATIENT End*				

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 4-2 - ORU^R01^ORU^R01 ABSTRACT MESSAGE SYNTAX					
Segment	**Name**	**Usage**	**Cardinality**	**Section**	**Description**
{	*ORDER_OBSERVATION Begin*	R	[1..*]		The order group is required and can repeat. This means that multiple ordered tests may be performed on a specimen.
[ORC]	Order Common	RE	[0..1]	5.8	The common order (ORC) segment identifies basic information about the order for testing of the specimen. This segment includes identifiers of the order, who placed the order, when it was placed, what action to take regarding the order, etc.
OBR	Observations Request	R	[1..1]	5.9	The observation request (OBR) segment is used to capture information about one test being performed on the specimen. Most importantly, the OBR identifies the type of testing to be performed on the specimen, and ties that information to the order for the testing.
[{NTE}]	Notes and Comments for OBR	RE	[0..*]		
[{	*TIMING_QTY Begin*	RE	[0..*]		
TQ1	Timing/Quantity	R	[1..1]		
[{TQ2}]	Timing/Quantity Order Sequence	O	[0..*]		
}]	*TIMING_QTY End*				
[CTD]	Contact Data	O	[0..1]		
[{	*OBSERVATION Begin*	CE	[0..*]		Multiple results may be associated with an order. There will always be a single OBX in the results group. Snapshot processing: Since the OBX segment in 2.5.1 does not contain a unique instance identifier, it is assumed that the repeating observation group will contain a complete set of observations (OBXs) associated with the OBR. Where a single OBX is being updated, all the OBXs related to the OBR must accompany the updated OBX, i.e., a full snapshot is sent. May be empty for OBR-25 Result statuses of "O," "I," "S" and "X"; otherwise, it is required.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 4-2 – ORU^R01^ORU^R01 ABSTRACT MESSAGE SYNTAX					
Segment	**Name**	**Usage**	**Cardinality**	**Section**	**Description**
OBX	Observation related to OBR	R	[1..1]	5.11	The observation/result (OBX) segment contains information regarding a single observation (analyze) result. This includes identification of the specific type of observation, the result for the observation, when the observation was made, etc.
[{NTE}]	Notes and Comments	RE	[0..*]		The notes and comment (NTE) segment may carry comments related to the result being reported in the OBX segment.
}]	*OBSERVATION End*				
[{FTI}]	Financial Transaction	O	[0..*]		
{[CTI]}	Clinical Trial Identification	O	[0..*]		
[{	*SPECIMEN Begin*	RE	[0..*]		The specimen group is not required in the ORU, but may be used to carry specimen information that is no longer contained in the OBR segment. It also provides a place for the specimen number. Each specimen group documents a single sample.
SPM	Specimen Information	R	[1..1]	5.12	The specimen information (SPM) segment describes the characteristics of a single sample. The SPM segment carries information regarding the type of specimen, where and how it was collected, who collected it, and some basic characteristics of the specimen.
[{OBX}]	Observation related to Specimen	O	[0..*]		
}]	*SPECIMEN End*				
}	*ORDER_ OBSERVATION End*				
}	*PATIENT_RESULT End*				
[DSC]	Continuation Pointer	X	[0..0]		Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

4.2 ACK^R01^ACK

TABLE 4-3 – ACK^R01^ACK ABSTRACT MESSAGE SYNTAX*					
Segment	**Name**	**Usage**	**Cardinality**	**Section**	**Description**
	Header Begin				
MSH	Message Header	R	[1..1]	5.2	The message header (MSH) segment contains information describing how to parse and process the message. This includes identification of message delimiters, sender, receiver, message type, timestamp, etc.
[{SFT}]	Software Segment	O	[0..*]		
MSA		R	[1..1]		
[{ ERR }]		C	[0..*]		Required when MSA-1 is not "AA" or "CA."

* See Section 7.2 for examples.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

This page intentionally left blank.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.Segment and Field Descriptions

This messaging guide provides notes for supported fields. The following format is used in this document for listing and defining message segments and fields. First, the message segment's use is defined, then a segment attribute table listing all fields defined in the segment is shown.

5.1 SEGMENT ATTRIBUTE TABLE ABBREVIATIONS

The following sections detail the structure of each segment attribute, including field length, data type, usage, cardinality, value set and description. The table below details the abbreviated terms used in the segment attribute tables, as well as the definitions associated with those abbreviations.

TABLE 5-1 – SEGMENT ATTRIBUTES

Abbreviation	Definition
Seq	Sequence of the elements as numbered in the HL7 segment.
Len	Maximum length of the element. Lengths are provided only for atomic data types. Lengths should be considered recommendations, not absolutes. The receiver can truncate fields, components, and sub-components that are longer than the recommended length. The receiver should continue to process a message even when a field, component, or sub-component length exceeds the maximum recommended length identified in this specification.
DT	Data type used by this profile for HL7 element.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-1 – SEGMENT ATTRIBUTES	
Abbreviation	**Definition**
Usage	Usage of the field for this profile. Indicates if the field, component, or subcomponent is required, optional, or conditional in the corresponding segment, field, or component. Legal values are: • R – Required. Must always be populated. • RE – Required, but can be empty. • O – Optional. May optionally be populated. • C – Conditional. Populated under specified conditions. • CE – Conditional, but can be empty. • X – Not used for this profile. Note: A required field in an optional segment does not mean the segment must be present in the message. Rather, if the segment is present, the required fields within that segment must be populated. This convention applies to required components of optional fields. If the field is populated, then the required components must be populated. The convention applies to required sub-components of optional components, as well. If a component is populated, then the required sub-components of that component must be populated.
Cardinality	Indicator of the minimum and maximum number of times the element may appear. • [0..0] Element never present. • [0..1] Element may be omitted and it can have at most one occurrence. • [1..1] Element must have exactly one occurrence. • [0..n] Element may be omitted or may repeat up to *n* times. • [1..n] Element must appear at least once, and may repeat up to *n* times. • [0..*] Element may be omitted or repeat for an unlimited number of times. • [1..*] Element must appear at least once, and may repeat unlimited number of times. • [m..n] Element must appear at least *m*, and at most, *n* times.
Value Set	The set of coded values to be used with the field. Note: Where a table constraint is indicated, or where HL7 Version 2.6 standards are pre-adopted, the constrained or specified HL7 table is included below the segment table.
HL7 Element Name	HL7 descriptor of the element in the segment.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-1 – SEGMENT ATTRIBUTES	
Abbreviation	**Definition**
Description/Comments	Context and usage for the element.

5.2 MSH – MESSAGE HEADER SEGMENT

The Message Header Segment (MSH) contains information describing how to parse and process the message. This includes identification of message delimiters, sender, receiver, message type, timestamp, etc.

TABLE 5-2 – MESSAGE HEADER SEGMENT (MSH)							
Seq	**Len**	**DT**	**Usage**	**Cardinality**	**Value Set**	**HL7 Element Name**	**Description/Comments**
1	1	ST	R	[1..1]		Field Separator	Character to be used as the field separator for the rest of the message. Literal value: '|' [ASCII (124)].
2	4	ST	R	[1..1]		Encoding Characters	Four characters, always appearing in the same order: |^~\&|. Literal value: '^~\&'.
3		HD	RE	[0..1]		Sending Application	Field that may be used to identify the sending application uniquely for messaging purposes. If populated, it will contain an OID that represents the sending application instance.
4		HD	R	[0..1]		Sending Facility	Field that uniquely identifies the facility that sends the message. If populated, it will contain an OID that represents the sending facility.
5		HD	RE	[0..1]		Receiving Application	Field that may be used to identify the receiving application uniquely for messaging purposes. If populated, it will contain an OID that represents the receiving application instance.
6		HD	RE	[0..1]		Receiving Facility	Field that uniquely identifies the facility that is to receive the message. If populated, it will contain an OID that represents the receiving facility.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-2 – MESSAGE HEADER SEGMENT (MSH)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
7		TS	R	[1..1]		Date/Time Of Message	Field containing the date/time the message was created by the sending system. Format: YYYYMMDDHHMMSS[.S[S[S[S]]]]+/-ZZZZ. Note that the time zone offset is required, and the minimum granularity is to the second, although more precise time stamps are allowed.
8		ST	O	[0..1]		Security	
9		MSG	R	[1..1]		Message Type	For the result message: 'ORU^R01^ORU_R01'. For the acknowledgement message: 'ACK^R01^ACK'.
10	999	ST	R	[1..1]		Message Control ID	String that uniquely identifies the message instance from the sending application.
11		PT	R	[1..1]		Processing ID	Field that may be used to indicate the intent for processing the message, such as "Testing," "Development," or "Production." For this message, the field will always contain |P|. The processing mode is understood to be "Current," if not explicitly sent in the message.
12		VID	R	[1..1]		Version ID	HL7 version number used to interpret format and content of the message. For this message, the version ID will always be 2.5.1
13		NM	O	[0..1]		Sequence Number	
14		ST	O	[0..1]		Continuation Pointer	
15	2	ID	CE	[1..1]	HL70155	Accept Acknowledgment Type	Condition Rule: For the Result message, must be valued with the literal value "AL." For the Accept Acknowledgment message, this field should be empty.
16	2	ID	CE	[0..1]	HL70155	Application Acknowledgment Type	Condition Rule: Must be left empty for the Accept Acknowledgment. Should be populated with a value from HL7 table 0155 for the Result message. Refer to *5.2.1 – HL7 Table 0155 – Accept/Application Acknowledgment Conditions* for valid values.
17		ID	O	[0..1]		Country Code	
18		ID	O	[0..*]		Character Set	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-2 - MESSAGE HEADER SEGMENT (MSH)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
19		CWE	O	[0..1]		Principal Language Of Message	
20		ID	O	[0..1]		Alternate Character Set Handling Scheme	
21		EI	R	[1..*]		Message Profile Identifier	Field used to reference or assert adherence to a message profile. Message profiles contain detailed explanations of grammar, syntax, and usage for a particular message or set of messages. This field is allowed to repeat. The profile ID for the profile defined in this guide should appear as a Repeat. Other profile IDs may appear in the field, as well, in cases where more constrained profiles are created from this profile. An OID for this profile is available once it is assigned. The OID appearing as the literal value is an example OID only. Literal Value: USLabReport^^2.16.840.1.113883.19.9.7^ISO

5.2.1 HL7 Table 0155 – Accept/Application Acknowledgment Conditions (Constrained from the Full HL7 Table)

TABLE 5-3 - HL7 TABLE 0155 - ACCEPT/APPLICATION ACKNOWLEDGMENT CONDITIONS (CONSTRAINED FROM FULL HL7 TABLE)			
Value	Description	Usage	Comment
AL	Always	R	
NE	Never	O	
ER	Error/reject conditions only	O	
SU	Successful completion only	O	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.3 MSA – ACKNOWLEDGEMENT SEGMENT

The Message Response Segment (MSA) contains the information sent as acknowledgment to the order message received by a Laboratory Information System.

TABLE 5-4 – ACKNOWLEDGEMENT SEGMENT (MSA)

Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	2	ID	R	[1..1]	HL70008	Acknowledgment Code	Acknowledgment code indicating receipt of message. (Refer to *HL7 Table 0008 - Acknowledgment Code* for valid values.)
2	999	ST	R	[1..1]		Message Control ID	Identifier that enables the sending system to associate this response with the message for which it is intended. This value will be the MSH.10 message control ID from the message being acknowledged.
3		ST	X	[0..0]		Text Message	Deprecated as of *HL7 Version 2.4*. See ERR segment.
4		NM	O	[0..1]		Expected Sequence Number	
5		ID	X	[0..0]		Delayed Acknowledgment Type	Deprecated as of *HL7 Version2.2*, and the detail was withdrawn and removed from the standard as of *HL7 Version 2.5*.
6		CWE	X	[0..0]		Error Condition	Deprecated as of *HL7 Version 2.4*. See ERR segment.

5.3.1 HL7 Table 0008 – Acknowledgement Code (Constrained from the Full HL7 Table)

TABLE 5-5 – HL7 TABLE 0008 – ACKNOWLEDGEMENT CODE (CONSTRAINED FROM FULL HL7 TABLE)

Value	Description	Usage	Comment
AA	Original mode: Application Accept - Enhanced mode: Application acknowledgment: Accept	O	
AE	Original mode: Application Error - Enhanced	O	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-5 – HL7 TABLE 0008 – ACKNOWLEDGEMENT CODE (CONSTRAINED FROM FULL HL7 TABLE)			
Value	Description	Usage	Comment
	mode: Application acknowledgment: Error		
AR	Original mode: Application Reject - Enhanced mode: Application acknowledgment: Reject	O	
CA	Enhanced mode: Accept acknowledgment: Commit Accept	R	
CE	Enhanced mode: Accept acknowledgment: Commit Error	R	
CR	Enhanced mode: Accept acknowledgment: Commit Reject	R	

5.4 ERR – ERROR SEGMENT

The ERR segment is used to add error comments to acknowledgment messages.

TABLE 5-6 – ERROR SEGMENT (ERR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1		ELD	X	[0..0]		Error Code and Location	Deprecated as of *HL7 Version 2.5*. See ERR-2 Error Location and ERR-3 HL7 Error Code fields.
2		ERL	O	[0..1]		Error Location	
3		CWE	R	[1..1]	HL70357	HL7 Error Code	Identifies the HL7 (communications) error code.
4	2	ID	R	[1..*]	HL70516	Severity	Identifies the severity of an application error. Knowing if something is Error, Warning or Information is intrinsic to how an application handles the content.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-6 - ERROR SEGMENT (ERR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
5		CWE	O	[0..1]		Application Error Code	
6		ST	O	[0..10]		Application Error Parameter	
7	65536	TX	R	[0..1]		Diagnostic Information	Information that may be used by help desk or other support personnel to diagnose a problem.
8		TX	X	[0..0]		User Message	Not supported.
9		IS	O	[0..*]		Inform Person Indicator	
10		CWE	X	[0..0]		Override Type	Not supported.
11		CWE	X	[0..0]		Override Reason Code	Not supported.
12		XTN	O	[0..*]		Help Desk Contact Point	

5.5 PID – PATIENT IDENTIFICATION SEGMENT

The Patient Identification Segment (PID) is used to provide basic demographics regarding the subject of the testing. The subject may be a person, animal, place, or thing.

TABLE 5-7 - PATIENT IDENTIFICATION SEGMENT (PID)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID – PID	For the first repeat of the PID segment, the sequence number shall be one (1), for the second repeat, the sequence number shall be two (2), etc.
2		CX	X	[0..0]		Patient ID	Deprecated as of *HL7 Version 2.3.1*. See PID-3 Patient Identifier List.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-7 – PATIENT IDENTIFICATION SEGMENT (PID)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
3		CX	R	[1..*]		Patient Identifier List	Field used to convey all types of patient/person identifiers. This includes social security numbers, driver's license numbers, medical record numbers, etc.
4		CX	X	[0..0]		Alternate Patient ID – PID	Deprecated as of *HL7 Version 2.3.1*. See PID-3.
5		XPN	R	[1..*]		Patient Name	Patient name or aliases. When the name of the patient is not known, a value must still be placed in this field since the field is required. In that case, HL7 recommends the following: |~^^^^^^U|. The "U" for the name type code in the second name indicates that it is unspecified. Since there may be no name components populated, this means there is no legal name, nor is there an alias. This guide will interpret this sequence to mean there is no patient name.
6		XPN	RE	[0..1]		Mother's Maiden Name	May be included for identification purposes. Name type code is constrained to the value "M."
7		TS	RE	[0..1]		Date/Time of Birth	Patient's date of birth. The time zone component is optional. Note that the granularity of the birth date may be important. For a newborn, birth date may be known down to the minute, while for adults it may be known only to the date. Birth date may be used by the lab to calculate an age for the patient, which may impact what normal ranges apply to particular test results.
8	20	IS	RE	[0..1]	HL70001	Administrative Sex	Patient's sex.
9		XPN	X	[0..0]		Patient Alias	Deprecated as of *HL7 Version 2.4*. See PID-5 - Patient Name.
10		CWE	RE	[0..*]	HL70005	Race	One or more codes that broadly refer to the patient's race(s).
11		XAD	O	[0..*]		Patient Address	
12		IS	X	[0..0]		County Code	Deprecated as of *HL7 Version 2.3*. See PID-11 - Patient Address, component 9 - County/Parish Code.
13		XTN	O	[0..*]		Phone Number – Home	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-7 – PATIENT IDENTIFICATION SEGMENT (PID)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
14		XTN	O	[0..*]		Phone Number – Business	
15		CWE	O	[0..*]		Primary Language	
16		CWE	O	[0..1]		Marital Status	
17		CWE	O	[0..1]		Religion	
18		CX	O	[0..1]		Patient Account Number	
19		ST	X	[0..0]		SSN Number – Patient	Deprecated as of *HL7 Version 2.3.1*. See PID-3 - Patient Identifier List.
20		DLN	X	[0..0]		Driver's License Number – Patient	Deprecated as of *HL7 Version 2.5*. See PID-3 - Patient Identifier List.
21		CX	O	[0..*]		Mother's Identifier	
22		CWE	RE	[0..*]	HL70189	Ethnic Group	
23		ST	O	[0..1]		Birth Place	
24		ID	O	[0..1]		Multiple Birth Indicator	
25		NM	O	[0..1]		Birth Order	
26		CWE	O	[0..*]		Citizenship	
27		CWE	O	[0..1]		Veterans Military Status	
28		CWE	X	[0..0]		Nationality	Deprecated as of *HL7 Version 2.4*. See PID-10 - Race, PID-22 - Ethnic group, and PID-26 - Citizenship.
29		TS	O	[0..1]		Patient Death Date and Time	
30		ID	O	[0..1]		Patient Death Indicator	
31	1	ID	RE	[0..1]	HL7136	Identity Unknown Indicator	Coded value indicating whether identity of patient is known or unknown.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-7 – PATIENT IDENTIFICATION SEGMENT (PID)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
32		IS	O	[0..*]		Identity Reliability Code	
33		TS	O	[0..1]		Last Update Date/Time	
34		HD	O	[0..1]		Last Update Facility	
35		CWE	X	[0..0]		Species Code	Not supported.
36		CWE	X	[0..0]		Breed Code	Not supported.
37		ST	X	[0..0]		Strain	Not supported.
38		CWE	X	[0..0]		Species Code	Not supported.
39		CWE	X	[0..0]		Tribal Citizenship	Not supported.

5.6 PV1 – PATIENT VISIT INFORMATION

This segment contains basic inpatient or outpatient encounter information.

TABLE 5-8 – PATIENT VISIT INFORMATION (PV1)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID - PV1	Sequence number of patient's visit being identified. Only one patient visit per message is permitted by this specification.
2	20	IS	R	[1..1]	HL70004	Patient Class	A gross identification of the classification of patient's visit
3		PL	C	[0..1]		Assigned Patient Location	Condition rule: Required if PV1-2 is "inpatient." Best practice is to populate this field as this may be the only geo-location provided for the patient in this message.
4		CWE	CE	[0..1]	UB04FL14	Admission Type	Condition rule: Required if PV1-2 is "inpatient."

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-8 - PATIENT VISIT INFORMATION (PV1)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
5		CX	O	[0..1]		Preadmit Number	
6		PL	O	[0..1]		Prior Patient Location	
7		XCN	O	[0..*]		Attending Doctor	
8		XCN	O	[0..*]		Referring Doctor	
9		XCN	O	[0..*]		Consulting Doctor	
10	20	IS	RE	[0..1]		Hospital Service	
11		PL	X	[0..0]		Temporary Location	Not supported.
12		IS	O	[0..1]		Preadmit Test Indicator	
13		IS	X	[0..0]		Re-admission Indicator	Not supported.
14		IS	X	[0..0]		Admit Source	Not supported.
15		IS	X	[0..0]		Ambulatory Status	Not supported.
16		IS	X	[0..0]		VIP Indicator	Not supported.
17		XCN	O	[0..*]		Admitting Doctor	
18		IS	O	[0..1]		Patient Type	
19		CX	O	[0..1]		Visit Number	
20		FC	O	[0..*]		Financial Class	
21		IS	X	[0..0]		Charge Price Indicator	Not supported.
22		IS	X	[0..0]		Courtesy Code	Not supported.
23		IS	X	[0..0]		Credit Rating	Not supported.
24		IS	X	[0..0]		Contract Code	Not supported.
25		DT	X	[0..0]		Contract Effective Date	Not supported.
26		NM	X	[0..0]		Contract Amount	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-8 – PATIENT VISIT INFORMATION (PV1)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
27		NM	X	[0..0]		Contract Period	Not supported.
28		IS	X	[0..0]		Interest Code	Not supported.
29		IS	X	[0..0]		Transfer to Bad Debt Code	Not supported.
30		DT	O	[0..1]		Transfer to Bad Debt Date	
31		IS	O	[0..1]		Bad Debt Agency Code	
32		NM	O	[0..1]		Bad Debt Transfer Amount	
33		NM	O	[0..1]		Bad Debt Recovery Amount	
34		IS	O	[0..1]		Delete Account Indicator	
35		DT	O	[0..1]		Delete Account Date	
36	20	IS	RE	[0..1]	HL70112	Discharge Disposition	Disposition of the patient at discharge or once the visit is completed, for example, "Discharged to Home/Self-Care", "Expired", "Left Against Medical Advice". Uses Uniform Billing codes.
37		DLD	O	[0..1]		Discharged to Location	
38		CWE	O	[0..1]		Diet Type	
39		IS	O	[0..1]		Servicing Facility	
40		IS	X	[0..0]		Bed Status	Deprecated as *of HL7 Version 2.3*. See PV1-3, component 5 - Location Status.
41		IS	O	[0..1]		Account Status	
42		PL	O	[0..1]		Pending Location	
43		PL	O	[0..1]		Prior Temporary Location	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-8 – PATIENT VISIT INFORMATION (PV1)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
44		TS	RE	[0..1]		Admit Date/Time	Date and time patient arrived for services
45		TS	RE	[0..*]		Discharge Date/Time	Date and time patient services ended
46		NM	O	[0..1]		Current Patient Balance	
47		NM	O	[0..1]		Total Charges	
48		NM	O	[0..1]		Total Adjustments	
49		NM	O	[0..1]		Total Payments	
50		CX	O	[0..1]		Alternate Visit ID	
51		IS	O	[0..1]		Visit Indicator	
52		XCN	X	[0..0]		Other Healthcare Provider	Deprecated as of *HL7 Version 2.3*. See PV1-3, component 5 - Location Status.

5.7 PV2 – PATIENT VISIT – ADDITIONAL INFORMATION SEGMENT

The PV2 segment is a continuation of information contained on the PV1 segment.

TABLE 5-9 – PATIENT VISIT – ADDITIONAL INFORMATION (PV2)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
1		PL	X	[0..0]		Prior Pending Location	Not supported.
2		CWE	X	[0..0]		Accommodation Code	Not supported.
3		CWE	RE	[0..1]		Admit Reason	A generalized explanation of why the patient needed services. Frequently used for chief complaint. Has no universally accepted value set.
4		CWE	X	[0..0]		Transfer Reason	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-9 - PATIENT VISIT - ADDITIONAL INFORMATION (PV2)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
5		ST	X	[0..0]		Patient Valuables	Not supported.
6		ST	X	[0..0]		Patient Valuables Location	Not supported.
7		IS	X	[0..0]		Visit User Code	Not supported.
8		TS	X	[0..0]		Expected Admit Date/Time	Not supported.
9		TS	X	[0..0]		Expected Discharge Date/Time	Not supported.
10		NM	X	[0..0]		Estimated Length of Inpatient Stay	Not supported.
11		NM	X	[0..0]		Actual Length of Inpatient Stay	Not supported.
12		ST	X	[0..0]		Visit Description	Not supported.
13		XCN	X	[0..0]		Referral Source Code	Not supported.
14		DT	X	[0..0]		Previous Service Date	Not supported.
15	1	ID	RE	[0..1]	HL70136	Employment Illness Related Indicator	Coded value indicating whether services are provided as a consequence of employment
16		IS	X	[0..0]		Purge Status Code	Not supported.
17		DT	X	[0..0]		Purge Status Date	Not supported.
18		IS	X	[0..0]		Special Program Code	Not supported.
19		ID	X	[0..0]		Retention Indicator	Not supported.
20		NM	X	[0..0]		Expected Number of Insurance Plans	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-9 - PATIENT VISIT - ADDITIONAL INFORMATION (PV2)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
21		IS	X	[0..0]		Visit Publicity Code	Not supported.
22		ID	X	[0..0]		Visit Protection Indicator	Not supported.
23		XON	RE	[0..*]		Clinic Organization Name	Name of the organization within a facility that provides patient care. Name shall be unique within any given facility.
24		IS	X	[0..0]		Patient Status Code	Not supported.
25		IS	X	[0..0]		Visit Priority Code	Not supported.
26		DT	X	[0..0]		Previous Treatment Date	Not supported.
27		IS	X	[0..0]		Expected Discharge Disposition	Not supported.
28		DT	X	[0..0]		Signature on File Date	Not supported.
29		DT	X	[0..0]		First Similar Illness Date	Not supported.
30		CWE	X	[0..0]		Patient Charge Adjustment Code	Not supported.
31		IS	X	[0..0]		Recurring Service Code	Not supported.
32		ID	X	[0..0]		Billing Media Code	Not supported.
33		TS	X	[0..0]		Expected Surgery Date and Time	Not supported.
34		ID	X	[0..0]		Military Partnership Code	Not supported.
35		ID	X	[0..0]		Military Non-Availability Code	Not supported.
36		ID	X	[0..0]		Newborn Baby Indicator	Not supported.
37		ID	X	[0..0]		Baby Detained Indicator	Not supported.
38		CWE	X	[0..0]		Mode of Arrival Code	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-9 - PATIENT VISIT - ADDITIONAL INFORMATION (PV2)							
Seq	Len	DT	Usage	Cardinality	Table	Data Element Name	Description/Comments
39		CWE	X	[0..0]		Recreational Drug Use Code	Not supported.
40		CWE	RE	[0..1]	HL70432	Admission Level of Care Code	A generalized identification of patient's acuity for the services received during the visit covered by this message
41		CWE	X	[0..0]		Precaution Code	Not supported.
42		CWE	X	[0..0]		Patient Condition Code	Not supported.
43		IS	X	[0..0]		Living Will Code	Not supported.
44		IS	X	[0..0]		Organ Donor Code	Not supported.
45		CWE	X	[0..0]		Advance Directive Code	Not supported.
46		DT	X	[0..0]		Patient Status Effective Date	Not supported.
47		TS	X	[0..0]		Expected LOA Return Date/Time	Not supported.
48		TS	X	[0..0]		Expected Pre-admission Testing Date/Time	Not supported.
49		IS	X	[0..0]		Notify Clergy Code	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.8 ORC – COMMON ORDER SEGMENT

The Common Order Segment (ORC) identifies basic information about the order for testing of the specimen. This segment includes identifiers for the order, who placed the order, when it was placed, what action to take regarding the order, etc.

TABLE 5-10 – COMMON ORDER SEGMENT (ORC)

Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	2	ID	R	[1..1]	HL70119	Order Control	Determiner of the function of the order segment. In the ORU^R01 this should be the literal value: 'RE'.
2		EI	CE	[0..1]		Placer Order Number	Same value as OBR-2 Placer Order Number, if OBR-2 Placer Order Number is populated.
3		EI	R	[1..1]		Filler Order Number	Same value as OBR-3 Filler Order Number (required).
4		EI	RE	[0..1]		Placer Group Number	The placer group number is used to identify a group of orders. In the laboratory setting this is commonly referred to as a requisition number.
5		ID	O	[0..1]		Order Status	
6		ID	O	[0..1]		Response Flag	
7		TQ	X	[0..0]		Quantity/Timing	Deprecated as of *HL7 Version 2.5*. See TQ1 and TQ2 segments.
8		EIP	O	[0..1]		Parent	
9		TS	O	[0..1]		Date/Time of Transaction	
10		XCN	O	[0..*]		Entered By	
11		XCN	O	[0..*]		Verified By	
12		XCN	O	[0..*]		Ordering Provider	
13		PL	O	[0..1]		Enterer's Location	
14		XTN	O	[0..*]		Call Back Phone Number	
15		TS	O	[0..1]		Order Effective Date/Time	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-10 - COMMON ORDER SEGMENT (ORC)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
16		CWE	O	[0..1]		Order Control Code Reason	
17		CWE	O	[0..1]		Entering Organization	
18		CWE	O	[0..1]		Entering Device	
19		XCN	O	[0..*]		Action By	
20		CWE	X	[0..0]		Advanced Beneficiary Notice Code	Not supported.
21		XON	O	[0..*]		Ordering Facility Name	
22		XAD	O	[0..*]		Ordering Facility Address	
23		XTN	O	[0..*]		Ordering Facility Phone Number	
24		XAD	O	[0..*]		Ordering Provider Address	
25		CWE	O	[0..1]		Order Status Modifier	
26		CWE	X	[0..0]		Advanced Beneficiary Notice Override Reason	Not supported.
27		TS	O	[0..1]		Filler's Expected Availability Date/Time	
28		CWE	O	[0..1]		Confidentiality Code	
29		CWE	O	[0..1]		Order Type	
30		CNE	O	[0..1]		Enterer Authorization Mode	
31		CWE	O	[0..1]		Parent Universal Service Identifier	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.9 OBR – OBSERVATION REQUEST SEGMENT

The Observation Request Segment (OBR) is used to capture information about one test being performed on the specimen. Most importantly, the OBR identifies the type of testing to be performed on the specimen and ties that information to the order for the testing.

TABLE 5-11 – OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID - OBR	Sequence number of one of multiple OBRs under one PID. For the first order transmitted, the sequence number shall be 1; for the second order, it shall be 2; and so on.
2		EI	RE	[0..1]		Placer Order Number	This identifier is assigned by the placer of the order being fulfilled by this result message. This identifier distinguishes the placer's order from all other orders created by the placer where an order is interpreted to be the testing identified in a single OBR segment. Normally, it is a type of system identifier assigned by the placer software application. The Placer Order Number and the Filler Order Number are essentially foreign keys exchanged between applications for uniquely identifying orders and the associated results across applications.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-11 - OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
3		EI	R	[1..1]		Filler Order Number	Order number associated with the Filling Application. This number is assigned to the test by the organization performing the test. This field should not contain the accession number or specimen identifier for a specimen. The specimen or accession identifier should be placed in SPM-2. The Filler Order Number identifies this order as distinct from all other orders being processed by this filler where an order is interpreted to be the testing identified in a single OBR segment.. Normally, this is a type of system identifier assigned by the filler software application. The Filler Order Number, along with the Placer Order Number, are essentially foreign keys exchanged between applications for uniquely identifying orders and the associated results across applications. In messages containing multiple OBRs, each OBR must be identified by a unique Filler Order Number. This is critical for making parent/child results relationships work properly. Microbiology cultures and sensitivities are linked in this fashion in this profile. See *Appendix A, Section A.4. Linking Parent and Child Results,* of this document for more information on linking parent/child results.
4		CWE	R	[1..1]		Universal Service Identifier	Identifier code for the requested observation/test/ battery. The Technical Committee recommends that additional language be added that notes that US Realm Universal Service Ids should be used, however, until such IDs are available use of local codes are permitted.
5		ID	X	[0..0]		Priority – OBR	Deprecated as of *HL7 Version 2.3*. See TQ1-9 Priority field.
6		TS	X	[0..0]		Requested Date/Time	Deprecated as of *HL7 Version 2.3*. See TQ1-8 Start date/time.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-11 - OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
7		TS	R	[1..1]		Observation Date/Time	For specimen-based observations, the date/time the specimen was collected. A minimum of year, month and day must be provided when the actual date/time is known. For unknown collection, date/time use "0000". If the SPM is sent, this field must contain the same value as the first component of SPM-17 Specimen Collection Date/Time. HL7 requires this field in an OBR in a result message.
8		TS	RE	[0..1]		Observation End Date/Time	For specimen-based observations where the specimen was collected over a period of time, this represents the end point in time when the specimen was collected. This field must contain the same value as the second component of SPM-17 Specimen Collection Date/Time.
9		CQ	X	[0..0]		Collection Volume	Deprecated as of *HL7 Version 2.5*. See SPM-12 Specimen Collection Amount.
10		XCN	O	[0..*]		Collector Identifier	
11		ID	O	[0..1]		Specimen Action Code	
12		CWE	O	[0..1]		Danger Code	
13		ST	O	[0..1]		Relevant Clinical Information	
14		TS	X	[0..0]		Specimen Received Date/Time	Deprecated as of *HL7 Version 2.5*. See SPM-18 Specimen Received Date/Time.
15		SPS	X	[0..0]		Specimen Source	Deprecated as of *HL7 Version 2.5*. See SPM-4 Specimen Type.
16		XCN	RE	[0..*]		Ordering Provider	Identifier of the provider who ordered the testing being performed. NPI may be used as the identifier.
17		XTN	O	[0..2]		Order Callback Phone Number	
18		ST	O	[0..1]		Placer Field 1	
19		ST	O	[0..1]		Placer Field 2	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-11 - OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
20		ST	O	[0..1]		Filler Field 1	
21		ST	O	[0..1]		Filler Field 2	
22		TS	R	[1..1]		Results Rpt/Status Chng - Date/Time	Required field in this message. Applies to the entire report. Receipt of a subsequent message with the same Filler Number and a different status in this field implies that processing may need to occur at the receiving application level to update a previous report.
23		MOC	O	[0..1]		Charge to Practice	
24	10	ID	RE	[0..1]	HL70074	Diagnostic Serv Sect ID	See *Appendix A, Section A.3*, of this document for more information regarding parent/child result reporting.
25	1	ID	R	[1..1]	HL70123	Result Status	
26		PRL	CE	[0..1]		Parent Result	Field that, together with OBR-29 Parent, allows this result to be linked to a specific OBX segment associated with another OBR segment. See *Appendix A, Section A.4. Linking Parent and Child Results,* of this document for more information on linking parent/child results. This field is required when linking child sensitivities to the parent culture.
27		TQ	X	[0..0]		Quantity/Timing	Deprecated as of *HL7 Version 2.5*. See TQ1 and TQ2 segments.
28		XCN	RE	[0..*]		Result Copies To	Identifier of a provider or organization that is to receive copies of these results. NPI may be used as the identifier.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-11 - OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
29		EIP	C	[0..1]		Parent	Used to link this OBR with a parent OBR. Commonly used with microbiology messages to link a susceptibility result with the parent culture that identified the organism. For this linkage to work properly, the Placer Order Number and the Filler Order Number must uniquely identify the specific parent OBR. This means that the same Filler Number cannot be used to identify multiple OBRs. See *Appendix A, Section A.4. Linking Parent and Child Results*, of this document for more information on linking parent/child results. Condition rule: This field is required if OBR-24 carries the value "MB" and OBR-4 indicates the ordered test is a culture and sensitivity.
30		ID	X	[0..0]		Transportation Mode	Not supported.
31		CWE	O	[0..*]		Reason for Study	
32		NDL	O	[0..1]		Principal Result Interpreter	
33		NDL	O	[0..*]		Assistant Result Interpreter	
34		NDL	O	[0..*]		Technician	
35		NDL	O	[0..*]		Transcriptionist	
36		TS	O	[0..1]		Scheduled Date/Time	
37		NM	X	[0..0]		Number of Sample Containers	Not supported. See SPM-26
38		CWE	X	[0..0]		Transport Logistics of Collected Sample	Not supported.
39		CWE	O	[0..*]		Collector's Comment	
40		CWE	X	[0..0]		Transport Arrangement Responsibility	Not supported.
41		ID	X	[0..0]		Transport Arranged	Not supported.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-11 - OBSERVATION REQUEST SEGMENT (OBR)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
42		ID	X	[0..0]		Escort Required	Not supported.
43		CWE	X	[0..0]		Planned Patient Transport Comment	Not supported.
44		CWE	O	[0..1]		Procedure Code	
45		CWE	O	[0..*]		Procedure Code Modifier	
46		CWE	O	[0..*]		Placer Supplemental Service Information	
47		CWE	O	[0..*]		Filler Supplemental Service Information	
48		CWE	O	[0..1]		Medically Necessary Duplicate Procedure Reason	
49		IS	O	[0..1]		Result Handling	
50		CWE	O	[0..1]		Parent Universal Service Identifier	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.10 TQ1 – TIMING/QUANTITY SEGMENT

The TQ1 segment is used to specify the timing of events and actions such as those that occur in order management and scheduling systems.

TABLE 5-12 – TIME/QUANTITY SEGMENT FOR ORDER GROUP (TQ1)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID - TQ1	Sequence number of the timing specification, the first of which shall be 1; the second of which shall be 2; and so on.
2		CQ	O	[0..1]		Quantity	
3		RPT	O	[0..*]		Repeat Pattern	
4		TM	O	[0..*]		Explicit Time	
5		CQ	O	[0..*]		Relative Time and Units	
6		CQ	O	[0..1]		Service Duration	
7		TS	RE	[0..1]		Start date/time	Field that may be specified by the requester, in which case it indicates the earliest date/time at which the services should be started. In many cases, however, the start date/time will be implied or will be defined by other fields in the service request record (*e.g.*, urgency - STAT). The filling service may record a value in this field after receipt of the service request.
8		TS	O	[0..1]		End date/time	
9		CWE	RE	[0..*]	HL70485	Priority	Urgency of the request. If this field is blank, the default is R (routine). Multiple priorities may be assigned to one order.
10		TX	O	[0..1]		Condition text	
11		TX	O	[0..1]		Text instruction	
12		ID	O	[0..1]		Conjunction	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-12 – TIME/QUANTITY SEGMENT FOR ORDER GROUP (TQ1)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
13		CQ	O	[0..1]		Occurrence duration	
14		NM	O	[0..1]		Total occurrence's	

5.11 OBX – OBSERVATION/RESULT SEGMENT

The Observation/Result Segment (OBX) contains information regarding a single observation. This includes identification of the specific type of observation, the result for the observation, when the observation was made, etc. Note that this segment is the balloting version of the OBX for 2.5.1, which contains new CLIA fields. It is included here to keep implementation of the OBX in line with the OBX in other lab result messages.

TABLE 5-13 – OBSERVATION/RESULT SEGMENT (OBX)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID – OBX	Sequence number of the OBX in relation to the OBR Observation segment to which it refers. The sequence number should increment by 1 for each OBX in the group.
2	3	ID	CE	[0..1]	HL70125 (see table in 5.11.1 below)	Value Type	This field identifies the data type used for OBX-5.. Conditional statement: if OBX-5 is populated, OBX-2 is required. See HL7 table 0125 below for what data types will be supported for this field (and OBX-5). Note that the field length has been extended to 3 characters to allow the 3 character data type codes from HL7 table 0125.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-13 - OBSERVATION/RESULT SEGMENT (OBX)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
3		CWE	R	[1..1]	LOINC, when available	Observation Identifier	Unique identifier for the type of observation. This field provides a code for the type of observation. OBX.3 in conjunction with OBX.4 Observation Sub-ID should uniquely identify this OBX from all other OBXs associated with this OBR. LOINC® is an HL7 approved code system and shall be used for the Observation Identifier as described in the appropriate HITSP Interoperability Specification. Use of LOINC codes for additional tests is strongly encouraged. See Section 6 for more details.
4	999	ST	C	[0..1]		Observation Sub-ID	Required if there is more than one OBX with the same OBX-3 (Observation Identifier) associated with the same OBR. Normally, this field is populated with a number, but text values may also be used.
5		Var	C	[0..1]		Observation Value	Field that documents each specific, allowed data type. Conditional statement: Either OBX-5 or OBX-8 (Abnormal flags) must be present in the message.
6		CWE	CE	[0..1]	UCUM, when applicable	Units	Required if the results have units. UCUM® is an HL7-approved code system, and shall be used for units as described in the appropriate HITSP Interoperability Specification. If a numeric value has no units of measure, the CWE data type allows an "NA" (Not Applicable) from HL70353 as the Identifier component. Conditional statement: If the data type in OBX 2 is "NM" or "SN," this field is required.
7	999	ST	RE	[0..1]		References Range	Interpretation range that applies to the value reported in OBX-5. It should provide enough information to understand the abnormal flags reported in OBX-8.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-13 - OBSERVATION/RESULT SEGMENT (OBX)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
8	20	IS	RE	[0..*]	HL70078	Abnormal Flags	Indicator of the normalcy of the result found in OBX-5. Cardinality indicates the possible need for multiple abnormal flags, as in the following example: *Example: Hemoglobin has a normal of 12-16* *Initial result: Specimen #1 reports a HGB = 15.9 (results normal)* *Subsequent result: Specimen #2 performed 1 hour later reports a HGB = 11.9 abnormality: (L) below low normal and a (D) significant change down (delta > 3).* *In this example, OBX-8 for Specimen #2 would be set to |L~D|.*
9		NM	O	[0..1]		Probability	
10		ID	O	[0..1]		Nature of Abnormal Test	
11	1	ID	R	[1..1]	HL7 0085	Observation Result Status	Status of the observation result.
12		TS	O	[0..1]		Effective Date of Reference Range	
13		ST	O	[0..1]		User-Defined Access Checks	
14		TS	RE	[0..1]		Date/Time of the Observation	For specimen based laboratory reporting, the specimen collection date and time. The date/time the testing was performed should be reported in OBX-19
15		CWE	O	[0..1]		Producer's Reference	
16		XCN	O	[0..*]		Responsible Observer	
17		CWE	O	[0..*]	HL7 V3 Observation Method	Observation Method	
18		EI	O	[0..*]		Equipment Instance Identifier	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-13 - OBSERVATION/RESULT SEGMENT (OBX)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
19		TS	RE	[0..1]		Date/Time of the Analysis	Time at which the testing was performed.
20		(TBD)	X	[0..0]		Reserved for harmonization with Version 2.6	Not supported.
21		(TBD)	X	[0..0]		Reserved for harmonization with Version 2.6	Not supported.
22		(TBD)	X	[0..0]		Reserved for harmonization with Version 2.6	Not supported.
23		XON	R	[1..1]		Performing Organization Name	The organization or service responsible for performing the service. When this field is null, the receiving system assumes that the observations were produced by the sending organization. The information for producer ID is recorded as an XON data type. In the US, the Medicare number of the producing service is suggested as the identifier (component 10). For laboratories, this field specifies the laboratory that produced the test result described in this OBX segment. It should be reported explicitly when the test results are produced at outside laboratories, for example. This information supports CLIA regulations in the US. For producing laboratories which are CLIA-certified, the CLIA identifier should be used for the organization identifier (component 10). CLIA requirements for laboratory identification may be addressed either by the receiving EHR having access to a lookup table that converts the CLIA ID in OBX-23 to fully populated demographics, or by fully populating OBX-23, -24,and -25.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-13 - OBSERVATION/RESULT SEGMENT (OBX)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
24		XAD	R	[1..1]		Performing Organization Address	Address of the laboratory that actually performed the test when used as a reference laboratory. CLIA requirements for laboratory identification may be addressed either by the receiving EHR having access to a lookup table that converts the CLIA ID in OBX-23 to fully populated demographics, or by fully populating OBX-23, -24, and -25.
25		XCN	RE	[0..1]		Performing Organization Medical Director	Name of the Medical Director of the reference laboratory. Required when OBX-24 indicates the performing lab is in a jurisdiction that requires this information. CLIA requirements for laboratory identification may be addressed either by the receiving EHR having access to a lookup table that converts the CLIA ID in OBX-23 to fully populated demographics, or by fully populating OBX-23,-24,and -25.

5.11.1 HL7 Table 0125 – Value Type (Constrained from the Full HL7 Table)

TABLE 5-14 - HL7 TABLE 0125 - VALUE TYPE (CONSTRAINED FROM FULL HL7 TABLE)			
Value	Description	Usage	Comment
AD	Address	X	
CE	Coded Entry	R	Data type to be used where it is important to communicate the coding system with the coded result being reported.
CF	Coded Element With Formatted Values	X	
CK	Composite ID With Check Digit	X	Withdrawn as of v 2.5
CN	Composite ID And Name	X	Withdrawn as of v 2.5

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-14 - HL7 TABLE 0125 - VALUE TYPE (CONSTRAINED FROM FULL HL7 TABLE)			
Value	**Description**	**Usage**	**Comment**
CP	Composite Price	X	
CWE	Coded with Exceptions	R	Data type to be used where it is important to communicate the coding system and coding system version with the coded result being reported.. Pre-adopted from Version 2.6
CX	Extended Composite ID With Check Digit	O	
DT	Date	O	
ED	Encapsulated Data	R	Field using the ED data type to allow communication of images, sound clips, XML documents, html markup, etc.
FT	Formatted Text (Display)	R	Field using the FT data type to carry a text result value. This is intended for display. The text may contain formatting escape sequences as described in the data types section. Numeric results and numeric results with units of measure should not be reported as text. These should be reported as NM or SN numeric results, with the units of measure in OBX-6.
MO	Money	X	
NM	Numeric	R	Field using the NM data type to carry a numeric result value. The only non-numeric characters allowed in this field are a leading plus (+) or minus (-) sign. The structured numeric (SN) data type should be used for conveying inequalities, ranges, ratios, etc. The units for the numeric value should be reported in OBX-6.
PN	Person Name	X	Withdrawn as of v 2.5
RP	Reference Pointer	R	Field using the RP data type to allow communication of pointers to images, sound clips, XML documents, html markup, etc. The RP data type is used when the object being pointed to is too large to transmit directly. This specification defines the mechanism for exchanging pointers to objects, but it does not address the details of applications actually accessing and retrieving the objects over a network. The most common scheme for passing a pointer is to use a Universal Resource Identifier (see `http://ietf.org/rfc/rfc2396.txt` for detailed definition). The general format of a URI is in the form:

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-14 – HL7 TABLE 0125 – VALUE TYPE (CONSTRAINED FROM FULL HL7 TABLE)			
Value	**Description**	**Usage**	**Comment**
			<scheme>://<authority><path>?<query>. The scheme and authority portions appear in the Application ID component, Universal ID subcomponent. The path and query portion of the URI appear in the Pointer component of the RP data type.
SN	Structured Numeric	R	Field using the SN data type to carry a structured numeric result value. Structured numeric include intervals (^0^-^1), ratios (^1^/^2 or ^1^:^2), inequalities (<^10), or categorical results (2^+). The units for the structured numeric value should be reported in OBX-6.
ST	String Data.	R	Field using the ST data type to carry a short text result value. Numeric results and numeric results with units of measure should not be reported as text. These should be reported as NM or SN numeric results, with the units of measure in OBX-6.
TM	Time	O	
TN	Telephone Number	X	Withdrawn as of v 2.5
TS	Time Stamp (Date & Time)	O	
TX	Text Data (Display)	R	Field using the TX data type to carry a text result value this is intended for display. Numeric results and numeric results with units of measure should not be reported as text. These should be reported as NM or SN numeric results, with the units of measure in OBX-6.
XAD	Extended Address	X	
XCN	Extended Composite Name And Number For Persons	X	
XON	Extended Composite Name And Number For Organizations	X	
XPN	Extended Person Name	X	
XTN	Extended Telecommunications Number	X	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

5.12 SPM – SPECIMEN SEGMENT

The Specimen Information Segment (SPM) describes the characteristics of a single sample. The SPM segment carries information regarding the type of specimen, where and how it was collected, who collected it and some basic characteristics of the specimen.

TABLE 5-15 – SPECIMEN SEGMENT (SPM)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
1	4	SI	R	[1..1]		Set ID – SPM	Number of instances of the SPM usage. For the first occurrence of the segment, the sequence number is |1|; for the second occurrence, the sequence number is |2|; etc.
2		EIP	RE	[0..1]		Specimen ID	Unique identifier for the specimen as referenced by the Placer application, the Filler application, or both.
3		EIP	O	[0..*]		Specimen Parent IDs	
4		CWE	R	[1..1]	HL70487 or SNOMED-CT	Specimen Type	Description of the precise nature of the entity that will be the source material for the observation.
5		CWE	O	[0..*]		Specimen Type Modifier	
6		CWE	O	[0..*]		Specimen Additives	
7		CWE	RE	[0..1]	HL70488 or SNOMED-CT	Specimen Collection Method	Method used to collect the specimen.
8		CWE	RE	[0..1]	SNOMED-CT	Specimen Source Site	Source from which the specimen was obtained. For environmental samples, this may describe the location of the source of the specimen. For biological samples it may represent the anatomical site from which the specimen was collected.
9		CWE	RE	[0..*]	SNOMED-CT	Specimen Source Site Modifier	Modifier or qualifier for the specimen source site (SPM-8). This allows use of post-coordinated terminologies for specimen source.
10		CWE	O	[0..1]		Specimen Collection Site	
11		CWE	O	[0..*]		Specimen Role	
12		CQ	RE	[0..1]	UCUM	Specimen Collection Amount	Amount of sample collected. This can be reported as a volume or a weight/mass.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE 5-15 – SPECIMEN SEGMENT (SPM)							
Seq	Len	DT	Usage	Cardinality	Value Set	HL7 Element Name	Description/Comments
13		NM	O	[0..1]		Grouped Specimen Count	
14		ST	O	[0..*]		Specimen Description	
15		CWE	O	[0..*]		Specimen Handling Code	
16		CWE	O	[0..*]		Specimen Risk Code	
17		DR	RE	[0..1]		Specimen Collection Date/Time	Time range over which the sample was collected, as opposed to the time the sample collection device was recovered.
18		TS	RE	[0..1]		Specimen Received Date/Time	Time the specimen is received at the diagnostic service. The actual time that is recorded is based on how specimen receipt is managed, and may correspond to the time the sample is logged in.
19		TS	O	[0..1]		Specimen Expiration Date/Time	
20		ID	O	[0..1]		Specimen Availability	
21		CWE	O	[0..*]		Specimen Reject Reason	
22		CWE	O	[0..1]		Specimen Quality	
23		CWE	O	[0..1]		Specimen Appropriateness	
24		CWE	O	[0..*]		Specimen Condition	
25		CQ	O	[0..1]		Specimen Current Quantity	
26		NM	O	[0..1]		Number of Specimen Containers	
27		CWE	O	[0..1]		Container Type	
28		CWE	O	[0..1]		Container Condition	
29		CWE	O	[0..1]		Specimen Child Role	

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

6. Code Systems and Value Sets

Successful message implementation requires that transmitted messages (message instances) contain valid values for coded fields. It is important to note that code sets are relatively dynamic and subject to change between publications of these implementation guides.

Every code value passed in a message instance is drawn from a code system that has a globally unique identifier, such as an OID. In general, the coded values allowed in a field (a) may be drawn from more than one code system, and (b) may be a subset of the codes from a given coding system. Combining (a) and (b) makes it possible for the allowed code value to be a combination of multiple subsets drawn from multiple coding systems. In most cases, only some of the codes defined in a code system are legal for use in a particular message.

The subsets of the codes that are legal for a particular field are identified by an HL7 construct known as a "value set." A value set is a collection of coded values drawn from code systems. Value sets serve to identify the specific set of coded values for the message from the universe of coded values across all coding systems. Value sets referenced in this guide are governed by *HITSP EHR Lab result Terminology Component C35* (Click here to see the document.). Below are the current requirements from that document. The actual content of any value set is subject to continual change, and the set of value sets may also change to reflect current needs.

The segment tables in previous sections identify the value set or coding system used for each supported field containing a coded value. Fields that use the data type CE or CWE require that messages include the code, drawn from *HL7 0396*, that uniquely defines the coding system, as well as the coded value itself. Some of these pre-coordinated value sets must be updated, or new ones created, as new needs are identified.

Value sets are identified by a unique identifier also, but this identifier is not transmitted in the message. The identifier or code for the coding system from which the value is derived is sent in the message. However, the value set identifier is useful and important when vocabulary items are modified or replaced.

6.1 VOCABULARY CONSTRAINTS

6.1.1 LOINC

TABLE 6-1 – LAB LOINC	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained	All LOINC lab result codes
Minimum attributes of the component:	HL7 value sets not established. Considered value sets should include: • HEDIS (Health plan Employer Data and Information Set) reported tests accounting for 95% of routine lab orders • Proposed value sets for micro and cytology codes per HITSP/C35. • Category A, B, & C bioterrorism agents/diseases • Public Health jurisdiction and Federal reportable disease conditions

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

6.1.2 SNOMED-CT

TABLE 6-2 - SNOMED-CT	
Code sets, vocabularies, terminologies and nomenclatures that need to be constrained:	SNOMED-CT
Minimum attributes of the component:	SNOMED-CT FDA SPL Problem List Subset SNOMED-CT Lab Test Findings Table, as applicable SNOMED-CT Organisms
Other Comments:	FDA SPL Problem List Subset available at `http://www.fda.gov/oc/datacouncil/term.html`

6.1.3 Unified Code for Units of Measure (UCUM)

The Unified Code for Units of Measure (UCUM) is a code system for units that is complete and free of all ambiguities, and assigns to each defined unit concise semantics. Each unit is defined using algebraic terms of other units. Those algebraic terms are also valid codes of the Unified Code for Units of Measure, http://aurora.regenstrief.org/UCUM/. Examples of most common laboratory units, and others, can be found at http://www.hl7.de/download/documents/ucum/ucum.html.

6.1.4 UB-04 Data Specification Manual

The value set (`UB04FL14`) used for PV1-4, Admit Type, is drawn from the official *UB-04 Data Specification Manual*, published July 2007, by the National Uniform Billing Committee (NUBC), and can be found at http://www.nubc.org. This coding system supersedes UB92 and is effective immediately (July, 2007).

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

7. Example Laboratory Result Messages

The examples in this section adhere to HL7 publishing requirements in that all persons and facilities are fictitious. They are taken from the *HL7 Version 3 Publishing Facilitator's Guide, Appendix D, Storyboard Names*.

Particular care has been given to demonstrating how OIDs would be used. The OIDs included are conformant with the HL7 OID Registry, which specifies that the HL7 root, 2.16.840.1.113883, is to be used plus the extension 19 to indicate that the OID is for example purposes only. This is followed by the normal HL7 ontology.

Emphasis has also been placed on demonstrating the use of standardized vocabularies together with local terminology. Use of the Specimen Segment (SPM) is included.

This profile will be registered with HL7 upon successful completion of balloting. In these examples, the fictitious OID 2.16.840.1.113883.19.9.7 has been applied for *USLabReport*.

The editors of this document want to acknowledge the help provided by the following people in assembling example messages in this section and *Appendix A*.

Margaret Marshburn SAIC - Science Applications International Corp.
Suzanne Nagami Kaiser Permanente
Greg Thomas Kaiser Permanente
Dr. Clement J. McDonald Regenstrief Institute for Health Care

7.1 MINIMAL MESSAGE WITH ACKNOWLEDGEMENT

The first example is a minimal message, including only those fields that are required. The subsequent examples include acknowledgements for a successful receipt, an error upon receipt and a rejection acknowledgement in response to the minimal message.

A commercial lab known as Reliable Labs, Inc., (OID # 2.16.840.1.113883.19.3.1) is sending the results of a Hemoglobin to the ordering facility, Lone Tree Clinic (OID# 2.16.840.1.113883.19.3.2).

Lone Tree Clinic has assigned the OID 2.16.840.1.113883.19.3.2.1 for its master patient index system administration. This is entered in the PID-3.4.

Note that although the PID-5 Patient Name may be empty, it has been populated in this example since the scenario would be unlikely to be one in which the name would not be available.

Lone Tree Clinic receives the message and, after inspecting the MSH-15, sends an Acknowledgement.

7.1.1 Example: Minimal Message

Reliable Labs, Inc., sends the following message:

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
MSH|^~\&||^2.16.840.1.113883.19.3.1^ISO||^2.16.840.1.113883.
    19.3.2^ISO|20070701132554-
    0400||ORU^R01^ORU_R01|20070701132554000008|P|2.5.1|||AL|
    NE|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||36363636^^^&2.16.840.1.113883.19.3.2&ISO^MR||Everywom
    an^Eve^E^^^^L ||19750410|F||2131-1^Other
    Race^HL70005|2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
OBR|1|20070701113255409^EHR^2.16.840.1.113883.19.3.2.3^ISO|2
    007070111132896^Lab^2.16.840.1.113883.19.3.1.6^ISO|14134
    -1^Hemoglobin^LN|||20070701123054-
    0400||||||20070701130030-
    0400||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI||||||20070701132554-
    0400||CH|F
OBX|1|NM|14134-1^Hemoglobin^LN||12.4|g/dL^^UCUM|12.0-
    16.0||||F||||||||||||Reliable Labs, Inc|3434 Industrial
    Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
```

7.1.2 Example: Successful Receipt Message

Lone Tree Clinic receives the message and, after inspecting the MSH-15, sends the following acknowledgement that receipt was successful:

```
MSH|^~\&||^2.16.840.1.113883.19.3.2^ISO||^2.16.840.1.113883.
    19.3.1^ISO|20070701151010-
    0400||ACK^R01^ACK|20070701151010000018|P|2.5.1||||||||U
    SLabReport^^2.16.840.1.113883.19.9.7^ISO
MSA|CA|20070701132554000008
```

7.1.3 Example: Error on Receipt Message

Reliable Labs, Inc., sends the following message

```
MSH|^~\&||^2.16.840.1.113883.19.3.1^ISO||^2.16.840.1.113883.
    19.3.2^ISO|20070701132554-
    0400||ORU^R01^ORU_R01|20070701132554000008|P|2.5.1|||AL|
    NE|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||36363636^^^&2.16.840.1.113883.19.3.2&ISO^MR||Everywom
    an^Eve^E^^^^L||19750410|F||2131-1^Other
    Race^HL70005|2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
OBX|1|NM|14134-1^Hemoglobin^LN||12.4|g/dL^^UCUM|12.0-
    16.0||||F||||||||||||Reliable Labs, Inc|3434 Industrial
    Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Lone Tree Clinic receives the message and discovers that the required segment OBR is missing. The following message is generated:

```
MSH|^~\&||^2.16.840.1.113883.19.3.2^ISO||^2.16.840.1.113883.
    19.3.1^ISO|20070701151010-
    0400||ACK^R01^ACK|20070701151010000018|P|2.5.1||||||||U
    SLabReport^^2.16.840.1.113883.19.9.7^ISO
MSA|CE|20070701132554000008
ERR|||100|E|||Missing required OBR segment
```

7.1.4 Example: Error on Receipt - Warning

Reliable Labs, Inc., sends the following message:

```
MSH|^~\&||^2.16.840.1.113883.19.3.1^ISO||^2.16.840.1.113883.
    19.3.2^ISO|20070701132554-
    0400||ORU^R01^ORU_R01|20070701132554000008|P|2.5.1|||AL|
    NE|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||36363636^^^&2.16.840.1.113883.19.3.2&ISO^MR||Everywom
    an^Eve^E^^^^L ||19750410|F||2131-1^Other
    Race^HL70005|2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
OBR|1|20070701113255409^EHR^2.16.840.1.113883.19.3.2.3^ISO|2
    007070111132896^Lab^2.16.840.1.113883.19.3.1.6^ISO|14134
    -100^Hemoglobin^LN|||20070701123054-
    0400||||||20070701130030-
    0400||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI||||||20070701132554-
    0400||CH|F
OBX|1|NM|14134-100^Hemoglobin^LN||12.4|g/dL^^UCUM|12.0-
    16.0||||F||||||||||||Reliable Labs, Inc|3434 Industrial
    Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
```

Lone Tree Clinic receives the message and discovers an invalid LOINC code in OBR-4. The following Acknowledgement message is generated indicating an Error and a Warning:

```
MSH|^~\&||^2.16.840.1.113883.19.3.2^ISO||^2.16.840.1.113883.
    19.3.1^ISO|20070701151010-
    0400||ACK^R01^ACK|20070701151010000018|P|2.5.1||||||||U
    SLabReport^^2.16.840.1.113883.19.9.7^ISO
MSA|CE|20070701132554000008
ERR|||103|W|||Invalid LOINC code
```

7.1.5 Example: Reject Receipt Message

Reliable Labs, Inc., sends the following message where the MSH-10 contains a processing code of "T":

```
MSH|^~\&||^2.16.840.1.113883.19.3.1^ISO||^2.16.840.1.113883.
    19.3.2^ISO|20070701132554-
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
0400||ACK^R01^ACK|20070701132554000008|P|2.5.1|||AL|NE||
|||USLabReport^^2.16.840.1.113883.19.9.7^ISO
```

(The remainder of the message is the same as that in *Section 7.1.1* above.)

Lone Tree Clinic rejects the message due to transmission of a Test message in the production environment and sends the following Reject Message:

```
MSH|^~\&||^2.16.840.1.113883.19.3.2^ISO||^2.16.840.1.113883.
19.3.1^ISO|20070701151010-
0400||ACK^R01^ACK|20070701151010000018|P|2.5.1||||||||U
SLabReport^^2.16.840.1.113883.19.9.7^ISO
MSA|CR|20070701132554000008
ERR|||202|E|||Test message sent in production environment
```

7.2 ELECTROLYTES LABORATORY RESULT MESSAGE

In this example, Reliable Labs, Inc., (OID # 2.16.840.1.113883.19.3.1) is sending the results of an electrolyte panel to the ordering facility, Lone Tree Clinic (OID# 2.16.840.1.113883.19.3.2).

7.2.1 Example: Electrolytes

This example demonstrates the use of most of the supported segments, with the exception of the NTE segment.

```
MSH|^~\&|Lab^2.16.840.1.113883.19.3.1.6^ISO|Reliable^2.16.84
0.1.113883.19.3.1^ISO|EHR^2.16.840.1.113883.19.3.2.3^ISO
|LoneTree^2.16.840.1.113883.19.3.2^ISO|20070701132554-
0400||ORU^R01^ORU_R01|20070701132554000008|D|2.5.1|||AL|
NE|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||00000111^^^&2.16.840.1.113883.19.3.2.1&ISO^MR||Everyw
oman^Eve^E^^^^L||19750410|F||2131-1^Other
Race^HL70005|2222 Home Street^^Ann
Arbor^MI^48103||^^^^^555^5552003
PV1|1|I|^^^^^N|1^Emergency^UB04FL14^^^^2008, v 2.0
PV2|||^Vomiting of three days duration
ORC|RE|222324^EHR^2.16.840.1.113883.19.3.2.3^ISO|434241^Lab^
2.16.840.1.113883.19.3.1.6^ISO
OBR|1|222324^EHR^2.16.840.1.113883.19.3.2.3^ISO|434241^Lab^2
.16.840.1.113883.19.3.1.6^ISO|34554-6^ELECTROLYTES HCFA
98 and VENOUS PH PANEL^LN^8000400^Electrolytes,
Plasma^99Lab|||20070701123054-
0400||||L|||20070701130030-
0400||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.16.840
.1.113883.19.4.6&ISO^^^^NPI||||||20070701132554-
0400||CH|F
TQ1|1||||||||S^Stat^HL70485
OBX|1|NM|2951-2^SODIUM^LN^230-
007^Na^99Lab||140|meq/L^^UCUM|137-
147||||F|||||^Beaker^Bill|||||||Reliable Labs, Inc|3434
Industrial Loop^^Ann
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
OBX|2|NM|22760-3^POTASSIUM^LN^230-
    006^K^99Lab||3.5|meq/L^^UCUM|3.4 -
    5.3||||F|||||^Beaker^Bill|||||||Reliable Labs, Inc|3434
    Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
OBX|3|NM|2075-0^CHLORIDE^LN^230-
    002^Cl^99Lab||107|meq/L^^UCUM|101 -
    111||||F|||||^Beaker^Bill|||||||Reliable Labs, Inc|3434
    Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
OBX|4|NM|2028-9^CARBON DIOXIDE, TOTAL^LN^230-003^CO2,
    Total^99Lab||140|meq/L^^UCUM|3.5 -
    5.5||||F|||||^Beaker^Bill|||||||Reliable Labs, Inc|3434
    Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
SPM|1|24243535^^2.16.840.1.113883.19.3.1.6^ISO||67922002^ser
    um^SCT|||VENIP^^HL70488|||||7^mL&milliliter&UCUM||||200
    70701124500-0400|20070701130030-0400
```

7.3 CREATININE CLEARANCE LABORATORY RESULT MESSAGE

In this example, Reliable Labs, Inc., (OID # 2.16.840.1.113883.19.3.1) is sending the results of a Creatinine Clearance to the ordering facility, Lone Tree Clinic (OID# 2.16.840.1.113883.19.3.2).

7.3.1 Example: Creatinine Clearance

This example demonstrates a result where 2 specimens are needed to process 1 order. The first OBX handles the Serum Creatinine; the second OBX, the Urine Creatinine; the third OBX, the urine Volume, the fourth, the calculated Creatinine Clearance, itself.

```
MSH|^~\&|Lab^2.16.840.1.113883.19.3.1.6^ISO|Reliable^2.16.84
    0.1.113883.19.3.1^ISO|EHR^2.16.840.1.113883.19.3.2.3^ISO
    |LoneTree^2.16.840.1.113883.19.3.2^ISO|20070701132554-
    0400||ORU^R01^ORU_R01|20070701132554000008|D|2.5.1|||AL|
    NE|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||00000111^^^&2.16.840.1.113883.19.3.2.1&ISO^MR||Everyw
    oman^Eve^E^^^^L||19750410|F||2131-1^Other
    Race^HL70005|2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
PV1|1|O
ORC|RE|222324^EHR^2.16.840.1.113883.19.3.2.3^ISO|434241^Lab^
    2.16.840.1.113883.19.3.1.6^ISO
OBR|1|222324^EHR^2.16.840.1.113883.19.3.2.3^ISO|434241^Lab^2
    .16.840.1.113883.19.3.1.6^ISO|34555-3^Creatinine
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
    clearance panel^LN|||20070701123054-
    0400||||L|||20070701130030-
    0400||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI||||||20070701132554-
    0400||CH|F
OBX|1|NM|2160-0^Creatinine^LN||1|mg/dL^^UCUM|0.6-1.1
    mg/dL||||F|||||^Beaker^Bill||||||Reliable Labs,
    Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
OBX|2|NM|20624-
    3^Creatinine^LN||100|mg/dL^^UCUM|||||F|||||^Beaker^Bill|
    ||||||Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
OBX|3|NM|3167-4^Specimen volume^LN||1.5|L^^UCUM|||||F
OBX|4|NM|2164-2^Creatinine renal
    clearance^LN||104|mL/min^^UCUM|75 to 115
    mL/min||||F|||||^Beaker^Bill||||||Reliable Labs,
    Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.84
    0.1.113883.19.4.6&ISO^^^^NPI
SPM|1|||SER^Serum^HL70487^^^^2.5.1
SPM|2|||UR^Urine^HL70487^^^^2.5.1||||||||1.5^L&&UCUM
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Appendix A. HL7 Reporting of Culture and Susceptibilities

A.1 INTRODUCTION

Parent-child relationships, such as culture and sensitivities, can be reported using the HL7 electronic messaging standard. However, this is an area where many vendors and large laboratories have augmented the standard to account for variations in the systems with which they work. This usually does not present a problem until these messages must be shared between systems (for instance, between laboratories and sub-contracted laboratories, or between laboratories and public health agencies).

Parent-child information such as culture and susceptibilities (*e.g.*, reporting of multi-resistant tuberculosis or drug-resistant gonococcus or pneumococcus) is a critical component of electronic, laboratory-based, public health reporting.

The approach described here is required for use in reporting microbiology results for this message profile.

A.2 TEMPLATE FOR CULTURE RESULTS

A template report for the initial identification of three organisms from a single stool culture is presented below. For each field (*i.e.*, the space between the pipes, "|"), a description of what should appear in that particular field is given, along with the segment-field number in parentheses (*e.g.*, OBR-3) for some of the fields. Note that these examples use the ORU^R01 message type.

```
MSH|…
PID|…
OBR|1| Placer number | Filler number | Identifier code for
    the requested test or panel of tests(OBR-4) |…
OBX|1|CE| Specific organism identifier (OBX-3) | Sub-id for
    the first organism (OBX-4) | Description of organism
    (OBX-5) |…
OBX|2|SN| Other identifier (OBX-3) | Sub-id for the first
    organism (OBX-4) | Observation on the organism (OBX-5) |…
OBX|3|CE| Specific organism identifier (OBX-3) | Sub-id for
    the second organism (OBX-4) | Description of organism
    (OBX-5) |…
OBX|4|SN| Other identifier (OBX-3) | Sub-id for the second
    organism (OBX-4) | Observation on the Organism (OBX-5) |…
OBX|5|CE| Specific organism identifier (OBX-3) | Sub-id for
    the third organism (OBX-4) | Description of organism
    (OBX-5) |…
OBX|6|SN| Other identifier (OBX-3) | Sub-id for the third
    organism (OBX-4) | Observation on the organism (OBX-5) |…
SPM|1| Specimen identifier for the specimen being tested|_
```

This report has the MSH (Message Header), the PID (Patient Identification Segment), a single OBR (Observation Request Segment), and six OBX (Observation/Results) segments, and a single SPM (Specimen Segment) Note that

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

the "Set ID" in the first field of each OBX is sequential, while the "Sub-ID" in the fourth field of each OBX is not sequential, but acts as a link for all of the OBX segments that are reporting information for a related observation. The "Sub-ID" field in the template above has the words "first," "second," and "third" in **bold** and highlighted in green. This is done to show that the identification of the first organism is the relating observation for the first two OBX segments (*e.g.*, Set-ID numbers 1 and 2). The identification of the second organism is the relating observation for the second two segments (*e.g.*, Set-ID numbers 3 and 4), and so on. An example using the template above is presented below.

A.2.1 Examples of Culture Results

In this example Reliable Labs, Inc., is sending preliminary results of a stool culture to the ordering facility, Lone Tree Clinic. Three pathogens have been identified: Campylobacter jejuni, Salmonella and Shigella.

This example shows the use of the Sub-ID in OBX-4 to connect related observations. The Sub-ID is shown in bolded letters and highlighted in green, as presented in the previous template. In this example, numbers are used for the Sub-ID. However, a text identifier such as "isolate1" could be used. The HL7 standard has defined the Sub-ID (*e.g.*, OBX-4) as a "string" data type. Thus, it can be either a number or text.

In this example, the information about colony counts in OBX segments with Set IDs 2, 4, and 6 is provided to show how the "Sub-ID" is used to relate the associated OBX segments to each other (*e.g.*, 1 and 2, 3 and 4, 5 and 6). Some laboratories may not have this additional information and would therefore transmit only the identification of the organisms (*e.g.*, OBX segments 1, 3 and 5).

Identified organisms should be reported as coded data instead of text data. Coded data enables machine processing of results. String data can normally only be interpreted by humans.

```
MSH|^~\&|Lab^2.16.840.1.113883.19.3.1.6^ISO|Reliable^2.16.840
    .1.113883.19.3.1^ISO|EHR^2.16.840.1.113883.19.3.2.3^ISO|L
    oneTree^2.16.840.1.113883.19.3.2^ISO|20070701132554-
    0400||ORU^R01^ORU_R01|20070701132554000008|D|2.5.1|||AL|N
    E|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||36363636^^^MPI&2.16.840.1.113883.19.3.2.1&ISO^MR||Ever
    ywoman^Eve^E^^^^L||19750410|F|||2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
PV1|1|O
PV2|||787.91^diarrhea, acute^I9C
ORC|RE|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700122^Lab^2
    .16.840.1.113883.19.3.1.6^ISO
OBR|1|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700122^Lab^2.
    16.840.1.113883.19.3.1.6^ISO|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|||20070701123054-
    0400||||||||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.
    16.840.1.113883.19.4.6&ISO^^^^NPI||||||20070701132554-
    0400||MB|P
OBX|1|CWE|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|1|66543000^Campyl
    obacter jejuni^SCT^^^^January
    2007||||||P|||||^Beaker^Bill|||||||Reliable Labs,
    Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|2|SN|564-5^COLONY COUNT:NUM:PT:XXX:QN:VC^LN|1|^10000^-
    ^90000|1^^UCUM|||||P|||||^Beaker^Bill|||||||Reliable
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
        Labs, Inc|3434 Industrial Loop^^Ann
        Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
        .1.113883.19.4.6&ISO^^^^NPI
    OBX|3|CWE|625-4^Bacteria
        identified:Prid:Pt:Stool:Nom:Culture^LN|2|302620005^Salmo
        nella group B phase 1 a-e^SCT^^^^January
        2007||||||P|||||^Beaker^Bill|||||||Reliable Labs,
        Inc|3434 Industrial Loop^^Ann
        Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
        .1.113883.19.4.6&ISO^^^^NPI
    OBX|4|SN|564-5^COLONY
        COUNT:NUM:PT:XXX:QN:VC^LN|2|>^100000|1^^UCUM|||||P|||||^B
        eaker^Bill|||||||Reliable Labs, Inc|3434 Industrial
        Loop^^Ann
        Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
        .1.113883.19.4.6&ISO^^^^NPI
    OBX|5|CWE|625-4^Bacteria
        identified:Prid:Pt:Stool:Nom:Culture^LN|3|77352002^Shigel
        la^SCT^^^^January
        2007||||||P|||||^Beaker^Bill|||||||Reliable Labs,
        Inc|3434 Industrial Loop^^Ann
        Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
        .1.113883.19.4.6&ISO^^^^NPI
    OBX|6|SN|564-5^COLONY
        COUNT:NUM:PT:XXX:QN:VC^LN|3|<^1000|1^^UCUM|||||P|||||^Bea
        ker^Bill|||||||Reliable Labs, Inc|3434 Industrial
        Loop^^Ann
        Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
        .1.113883.19.4.6&ISO^^^^NPI
    SPM|1|38294523&&2.16.840.1.114222.4.3.2&ISO^9876543&&2.16.840
        .1.114222.4.3.2&ISO||119339001^Stool specimen^SCT
```

A.3 TEMPLATE FOR CULTURE AND SUSCEPTIBILITY RESULTS

The template and example in the *Template for Culture Results* section of this appendix describe a report for a culture. The following template shows how antimicrobial susceptibility results are reported for the culture described in that section. The connection of the culture to the susceptibilities is a "parent-child" relationship, where the culture is the parent result and the susceptibilities are the child results. This means that there can be many child results for a single parent result. In other words, there can be multiple OBR child segments for the single OBR parent segment presented in the Template for Culture Results section of this appendix. The template for the report containing the culture and susceptibilities appears below. The titles in Italics are given to highlight the individual parent and child segments and are not found in an actual HL7 message transmission. It is important to note that in each of the OBR child segments, there is a pointer back to the parent result. This pointer is found in OBR-26 ("Parent Result") and in OBR-29 ("Parent Number").

Message Header and Patient Identification Segment for the Parent-Child Message

```
    MSH|…
    PID|…
```

Parent OBR Segment

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

OBR|1| Placer number (OBR-2) | Filler order number (OBR-3) | Identifier code for the requested test or panel of tests (OBR-4) |…

Parent OBX Segments for First Organism Identified

OBX|1|CE| Specific organism identifier (OBX-3) | Sub-id for the **first** organism (OBX-4) | Description of organism (OBX-5) |…

OBX|2|SN| Other identifier (OBX-3) | Sub-id for the **first** organism (OBX-4) | Observation on the organism (OBX-5) |…

Parent OBX Segments for Second Organism Identified

OBX|3|CE| Specific organism identifier (OBX-3) | Sub-id for the **second** organism (OBX-4) | Description of organism (OBX-5) |…

OBX|4|SN| Other identifier (OBX-3) | Sub-id for the **second** organism (OBX-4) | Observation on the Organism (OBX-5) |…

Parent OBX Segments for Third Organism Identified

OBX|5|CE| Specific organism identifier (OBX-3) | Sub-id for the **third** organism (OBX-4) | Description of organism (OBX-5) |…

OBX|6|SN| Other identifier (OBX-3) | Sub-id for the **third** organism (OBX-4) | Observation on the organism (OBX-5) |…

Child OBR for First Organism identified

OBR|2| Placer number (OBR-2)| Filler order number (OBR-3) | Identifier code for the requested test or panel of tests (OBR-4) ||||||||||||||||||||| A **pointer** back to the parent OBX segment that contained the identification of the **first** organism, see below for description of "Pointers" (OBR-26) ||| Parent Filler order number (OBR-29) |...

Child OBX Segments for Susceptibilities of First Organism Identified

OBX|1|CE|Specific susceptibility identifier for first antimicrobial (OBX-3) || Susceptibility finding (OBX-5) ||| Susceptibility interpretation (OBX-8) |…

OBX|2|CE|Specific susceptibility identifier for second antimicrobial (OBX-3) || Susceptibility finding (OBX-5) ||| Susceptibility interpretation (OBX-8) |…

OBX|3|CE|Specific susceptibility identifier for third antimicrobial (OBX-3) || Susceptibility finding (OBX-5) ||| Susceptibility interpretation (OBX-8) |…

Child OBR Segment for Susceptibilities of Second Organism Identified

OBR|3| Placer number (OBR-2)| Filler order number (OBR-3) | Identifier code for the requested test or panel of tests

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
(OBR-4) |||||||||||||||||||||| A pointer back to the parent
OBX segment that contained the identification of the second
organism, see below for description of "Pointers" (OBR-26)
||| Parent Filler order number (OBR-29) |...
```

Child OBX Segments for Susceptibilities of Second Organism Identified

```
OBX|1|CE|Specific susceptibility identifier for first
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…

OBX|2|CE|Specific susceptibility identifier for second
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…

OBX|3|CE|Specific susceptibility identifier for third
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…
```

Child OBR Segment for Susceptibilities of Third Organism Identified

```
OBR|3| Placer number (OBR-2)| Filler order number (OBR-3) |
Identifier code for the requested test or panel of tests
(OBR-4) |||||||||||||||||||||| A pointer back to the parent
OBX segment that contained the identification of the third
organism, see below for description of "Pointers" (OBR-26)
||| Parent Filler order number (OBR-29) |...
```

Child OBX Segments for Susceptibilities of Third Organism Identified

```
OBX|1|CE|Specific susceptibility identifier for first
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…

OBX|2|CE|Specific susceptibility identifier for second
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…

OBX|3|CE|Specific susceptibility identifier for third
antimicrobial (OBX-3) || Susceptibility finding (OBX-5) |||
Susceptibility interpretation (OBX-8) |…
```

SPM Segment

```
SPM|1| Specimen identifier for the specimen being tested|…
```

A.3.1 Examples of Culture and Susceptibility Results

Using the template above, this example shows a report of three pathogens identified from a stool specimen with their respective antimicrobial susceptibility tests.

```
MSH|^~\&|Lab^2.16.840.1.113883.19.3.1.6^ISO|Reliable^2.16.840
    .1.113883.19.3.1^ISO|EHR^2.16.840.1.113883.19.3.2.3^ISO|L
    oneTree^2.16.840.1.113883.19.3.2^ISO|20070708123554-
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
    0400||ORU^R01^ORU_R01|20070701123554000008|D|2.5.1|||AL|N
    E|||||USLabReport^^2.16.840.1.113883.19.9.7^ISO
PID|1||36363636^^^MPI&2.16.840.1.113883.19.3.2.1&ISO^MR||Ever
    ywoman^Eve^E^^^^L||19750410|F|||2222 Home Street^^Ann
    Arbor^MI^48103||^^^^^555^5552003
PV1|1|O
PV2|||787.91^diarrhea, acute^I9C
ORC|RE|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700122^Lab^2
    .16.840.1.113883.19.3.1.6^ISO
OBR|1|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700122^Lab^2.
    16.840.1.113883.19.3.1.6^ISO|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|||20070701123054-
    0400||||||||000100^Hippocrates^Harold^H^^Dr^MD^^NPPES&2.
    16.840.1.113883.19.4.6&ISO^^^^NPI|||||20070701132554-
    0400||MB|F
OBX|1|CWE|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|1|66543000^Campyl
    obacter
    jejuni^SCT^^^^xxx||||||P|||||^Beaker^Bill|||||||Reliable
    Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|2|SN|564-5^COLONY COUNT:NUM:PT:XXX:QN:VC^LN|1|^10000^-
    ^90000|1^^UCUM|||||P|||||^Beaker^Bill|||||||Reliable
    Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|3|CWE|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|2|302620005^Salmo
    nella group B phase 1 a-
    e^SCT^^^^xxx||||||P|||||^Beaker^Bill|||||||Reliable Labs,
    Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|4|SN|564-5^COLONY
    COUNT:NUM:PT:XXX:QN:VC^LN|2|>^100000|1^^UCUM|||||P|||||^B
    eaker^Bill|||||||Reliable Labs, Inc|3434 Industrial
    Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|5|CWE|625-4^Bacteria
    identified:Prid:Pt:Stool:Nom:Culture^LN|3|77352002^Shigel
    la^SCT^^^^xxx||||||P|||||^Beaker^Bill|||||||Reliable
    Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|6|SN|564-5^COLONY
    COUNT:NUM:PT:XXX:QN:VC^LN|3|<^1000|1^^UCUM|||||P|||||^Bea
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
    ker^Bill||||||Reliable Labs, Inc|3434 Industrial
    Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPISPM|1|38294523&EHR&2.16.840.1.
    113883.19.3.2.3&ISO^9876543&Lab&2.16.840.1.113883.19.3.1.
    6&ISO||119339001^Stool specimen^SCT
OBR|2|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700123^Lab^2.
    16.840.1.113883.19.3.1.6^ISO|50545-3^Bacterial
    susceptibility panel:-
    :Pt:Isolate:OrdQn:MIC^LN|||20070701123054-
    0400||||||||||||||200502081402|||F|625-4&Bacteria
    identified:Prid:Pt:Stool:Nom:Culture&LN^1^Campylobacter
    jejuni|||23456&EHR&2.16.840.1.113883.19.3.2.3&ISO^9700122
    &Lab&2.16.840.1.113883.19.3.1.6^&ISO|
OBX|1|SN|6979-9^AMPICILLIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||<^0.06|ug/mL^^UCUM||S|||P|||||^Beaker^Bill|||||
    ||Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|2|SN|7016-9^GENTAMICIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|ug/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|3|SN|7002-9^CIPROFLOXACIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|ug/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBR|3|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|Lab^2.16.840.1
    .113883.19.3.1.6|50545-3^Bacterial susceptibility panel:-
    :Pt:Isolate:OrdQn:MIC^LN|||20070701123054-
    0400||||||||||||||200502081402|||F|625-4&Bacteria
    identified:Prid:Pt:Stool:Nom:Culture&LN^2^Salmonella
    group B phase 1 a-
    e|||23456&EHR&2.16.840.1.114222.4.3.2&ISO^9700122&LAB&2.1
    6.840.1.114222.4.3.2&ISO
OBX|1|SN|6979-9^AMPICILLIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||<^0.06|ug/mL^^UCUM||S|||P|||||^Beaker^Bill|||||
    ||Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|2|SN|7016-9^GENTAMICIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|ug/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
```


Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

```
OBX|3|SN|7002-9^CIPROFLOXACIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|ug/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBR|4|23456^EHR^2.16.840.1.113883.19.3.2.3^ISO|9700125^Lab^2.
    16.840.1.113883.19.3.1.6^ISO|50545-3^Bacterial
    susceptibility panel:-
    :Pt:Isolate:OrdQn:MIC^LN|||20070701123054-
    0400||||||||||||||200502081402|||F|625-4&Bacteria
    identified:Prid:Pt:Stool:Nom:Culture&LN^2^Shigella|||2345
    6&EHR&2.16.840.1.113883.19.3.2.3&ISO^9700122&Lab&2.16.840
    .1.113883.19.3.1.6^&ISO
OBX|1|SN|6979-9^AMPICILLIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||<^0.06|µg/mL^^UCUM||S|||P|||||^Beaker^Bill|||||
    ||Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|2|SN|7016-9^GENTAMICIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|µg/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
OBX|3|SN|7002-9^CIPROFLOXACIN:SUSC:PT:ISLT:ORDQN:GRADIENT
    STRIP^LN||^0.05|µg/mL^^UCUM||S|||P|||||^Beaker^Bill||||||
    |Reliable Labs, Inc|3434 Industrial Loop^^Ann
    Arbor^MI^48103^^B|9876543^Slide^Stan^S^^^^^NPPES&2.16.840
    .1.113883.19.4.6&ISO^^^^NPI
```

A.4 LINKING PARENT AND CHILD RESULTS

The previous example uses the information in OBR-26 as a pointer back to the parent OBX where the culture result is reported. OBR-26 has three components. The three components of OBR-26 are essentially the OBX-3, OBX-4 and part of the OBX-5 from the parent OBX segment. The pointer back to the parent requires only the first two components. The third component is intended to provide additional information that may be useful, but not necessary. This allows a lengthy result in the parent OBX-5 (*e.g.*, a paragraph describing pathology results) to be truncated or not sent at all.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

Appendix B. Clinical Laboratory Improvements Amendment Considerations, US Realm Only

In the United States, Clinical Laboratory testing of human specimens is regulated by the Clinical Laboratory Improvements Amendments of 1988 (CLIA). Several sections of the regulations implementing CLIA impact how electronic laboratory is formatted for the US Realm and these changes are outlined in this Appendix. Impacted areas include mandatory reporting requirements, report retention and display, and those authorized to receive a report. Specifics on the CLIA Regulation are found at http://wwwn.cdc.gov/clia/regs/toc.aspx.

B.1 MANDATORY REPORTING REQUIREMENTS

Section 493.1291 of the CLIA Regulations defines items that must appear on a clinical laboratory report in the US (http://wwwn.cdc.gov/clia/regs/subpart_k.aspx#493.1291). Interpretative Guidelines on the elements required in a report may be found at http://www.cms.hhs.gov/CLIA/downloads/apcsubk2.pdf. Specific report fields impacted include the following:

TABLE B-1 – MANDATORY REPORTING REQUIREMENTS

Segment	Field	CLIA Impact
PID-3	Patient Identifier List	A unique patient identification number is required
PID-5	Patient Name	Positive patient identification required. If the patient's name is known, this must be that name. If it is not known, a unique patient identifier must be assigned.
OBX-3	Observation Identifier[3]	Unique identification of the test performed is required. LOINC® is an HL7-approved code system and shall be used for the Observation Identifier as described in the appropriate HITSP Interoperability Specification. Use of LOINC codes for additional tests is strongly encouraged. See *Section 6* for more details. Addition of a local laboratory code is allowed. For certain tests CLIA requires additional information: Laboratories using manufacturer's instruments, kits or test systems labeled for "investigational use only" or "research use

[3] While CLIA requires a laboratory to maintain positive identification of a specimen reporting that information as part of the result is not required.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE B-1 – MANDATORY REPORTING REQUIREMENTS		
Segment	**Field**	**CLIA Impact**
		only" must clearly state that the test results are not to be used for treatment or diagnostic purposes. If results of such tests are being reported without a disclaimer statement, or are being used by the provider for patient care, they are in the same category as in-house developed tests and the laboratory must establish performance specifications in accordance with §493.1253. The disclaimer for Analyte Specific Reagents (ASR) should state, "This test was developed and its performance characteristics determined by (Laboratory Name). It has not been cleared or approved by the U.S. Food and Drug Administration." The ASR disclaimer on the test report is required by the FDA under *21 CFR, Part 809.30, "Restrictions on the sale, distribution and use of analyte -specific reagents."*
OBX-5	Observation Value	The laboratory result is required. No regulatory requirements are specified, outside of readability, regarding result appearance.
OBX-6	Units	Units, if required, or an interpretation must be given. For tests such as genetic screens the interpretation may actually be the test result.
OBX-7	Reference Range	Required.
OBX-8	Abnormal flag.	Use is not required, but a laboratory may use this field as part of its interpretation guidance. If reported, it should be displayed by an EHR.
OBX-11	Observation Result Status	Used to reflect CLIA required conditions such as specimen acceptability, result corrections, cancellations as well as report status (§493.1291 (c)(7) and (k)(1,2). See SPM- 21, and 22 below.
OBX-19	Date/Time of Analysis	Use this field to meet the requirement for test report time.
OBX-23, 24, 25	Laboratory Identification Fields	The identification of the performing laboratory is required. Populating with the CLIA ID Number in OBX-23 meets the requirement if this receiving EHR has access to a look-up table that will convert the CLIA ID number to full demographics comprising OBX-23 ,Performing Organization Name; OBX-24, Performing Organization Address; and OBX-25, Performing Organization Medical Director. If the CLIA ID Number is not used, all demographic fields (OBX-23, OBX-24 and OBX-25) must be populated with appropriate information.
SPM-4	Specimen Type	Reporting requirements call for the specimen source, which equates to the Specimen Type in the SPM segment.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

TABLE B-1 – MANDATORY REPORTING REQUIREMENTS		
Segment	**Field**	**CLIA Impact**
SPM-21	Specimen Reject Reason	Use this field in connection with OBX-11 if a test is cancelled for specimen related reason. SNOMED-CT is the recommenced terminology.
SPM-22	Specimen Quality	Use this field to provide a coded version for Specimen Description. For Electronic Health Records, it is preferred that this field be used in place of, or in connection with, SPM-14. SNOMED-CT is the recommenced terminology.

B.2 REPORT RETENTION

While this section is not to be construed as legal advice, the electronic message from a performing laboratory is presumed to be the legal report of the tests performed. Hence, the receiver must save the content of the message for the same time period as required for any other legal document.

An EHR receiving electronic laboratory reports should be able to reconstruct the content of original message and display it upon user request.

The EHR, however, has no obligations under CLIA to display or verify the required reporting elements noted above. That is the obligation of the sending laboratory. Likewise, the EHR should display as a clinical report only those portions of an electronic laboratory report that are deemed important by the using provider.

B.3 AUTHORIZED PARTIES

Local laws, generally at the State level, govern who is authorized to receive laboratory reports. CLIA restricts the availability of those authorized to receive laboratory reports to just those approved at the local level and sets no national standards. Testing laboratories may not report results to unauthorized parties under CLIA.

Testing laboratories either have a trusted relationship with the ordering party or presume that the ordering party is authorized to receive results. However, testing laboratories need not have knowledge of the appropriateness of others requested to receive results, such as "Copy to" recipients. To maintain CLIA compliance, a US testing laboratory may choose to restrict its reports to only those recipients known to be authorized. Hence, copies of a result need not be sent by a testing laboratory. Note that CLIA places no restrictions on the receiver of a laboratory report regarding its retransmission of the report to others.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.

This page intentionally left blank.

Copyright © 2007-2008 Health Level Seven International. All Rights Reserved.